QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

bitbox02-firmware-en @ ae548ef4e40f · BitBox (Shift Crypto) — public repository (passive review)

Organization	QSI
Report date	2026-10-29 05:31 UTC
Report version	1
Audit run	5e18f914-8dcb-4f3c-b1a4-538f892ea149 · Document audit · 2026-10-29 05:31 UTC
Analyzer engine	qsi-engine 1.2.0
Risk profile	Crypto exchange, custodian or wallet
Content SHA-256	`0f265adc21c8c990d44fbf098fcee7eebde1b04c28396bf07b7094b1e54e848b`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

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This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

17	0	0	11	0
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	1
Known vulnerable component	4
Confirmed configuration error	0
Potential cryptographic risk	0
Long-term post-quantum risk	10
Improvement recommendation	1
Insufficient data	1

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-06382	Medium	Transaction signing for 3 chain(s) relies on elliptic-curve signatures confirmed by the analyst	56.8
QSI-2026-06373	Medium	Quantum-vulnerable ECDSA: firmware and boot verification (root of trust) (3 file(s)) confirmed by the analyst	43.4
QSI-2026-06379	Medium	Quantum-vulnerable X25519: key wrapping and data encryption at rest (1 file(s)) confirmed by the analyst	43.4
QSI-2026-06375	Medium	Quantum-vulnerable X25519: channel key exchange (TLS, Noise, handshakes) (3 file(s)) confirmed by the analyst	41.8
QSI-2026-06376	Medium	Quantum-vulnerable ECDSA: transaction and message signing (50 file(s)) confirmed by the analyst	41.8

Mosca's inequality (x + y > z): x = 10, y = 5, z = 8 → migration would finish too late under this scenario. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Analyst review: 12 confirmed, 5 not applicable, 0 dismissed as false positives.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography. Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Scope and boundaries

Mode: Document audit

Repository: https://github.com/BitBoxSwiss/bitbox02-firmware @ ae548ef4e40f548a17542ba457f138db4fa0069d — evidence links open the lines at this commit.

Supplied inputs

721 files, 11803951 bytes in total, 16 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.2.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Crypto exchange, custodian or wallet" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
chain_map	checked	3	messages/btc.proto: Bitcoin; messages/cardano.proto: Cardano; messages/eth.proto: Ethereum and EVM chains
source_code	checked	664	src/atecc/atecc.c: 795 lines scanned with 16 rule patterns and 123 registry patterns; src/atecc/atecc.h: 38 lines scanned with 16 rule patterns and 123 registry patterns; src/bootloader/boot_args.h: 69 lines scanned with 16 rule patterns and 123 registry patterns
dependencies	checked	40	src/rust/Cargo.lock: 135 dependencies parsed (135 pinned); src/rust/Cargo.toml: 18 dependencies parsed (0 pinned); src/rust/bins/bitbox03-boot0/Cargo.toml: 2 dependencies parsed (0 pinned)
secrets	checked	8	src/rust/bitbox-secp256k1/src/lib.rs: value(s) at line(s) 375 (test_dleq), 407 (test_secp256k1_sign), 441 (test_secp256k1_sign_zero_host_nonce_differs_from_no_host_nonce_sign), 456 (test_secp256k1_sign_no_host_nonce_deterministic), 491 (test_secp256k1_nonce_commit) are declared inside a fixture and treated as test data, although the file itself is not test code; src/rust/bitbox-secp256k1/src/lib.rs: 5 redacted value(s) not reported (comments, test modules, fixture declarations, placeholders, values the source annotates as published, or hex values not bound to a secret name); src/rust/bitbox02/src/securechip/imp_fake.rs: 1 redacted value(s) not reported (comments, test modules, fixture declarations, placeholders, values the source annotates as published, or hex values not bound to a secret name)
osv_advisories	checked	1	135 pinned packages queried at OSV.dev; 6 affected

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 708 (input files: C)

Architecture overview

Static review of the source tree bitbox02-firmware-en at ae548ef4e40f548a17542ba457f138db4fa0069d. 721 files analysed; of 1013 files, 78 are test, fuzzing, benchmark or example code and were left out, 214 are not read by any analyzer, 0 exceed the per-file limit and 0 did not fit the total limit. 132 files come from directories a .gitmodules pins and are read as shipped source rather than skipped as a dependency cache. No requests were sent to the owner's systems.

Overall security posture

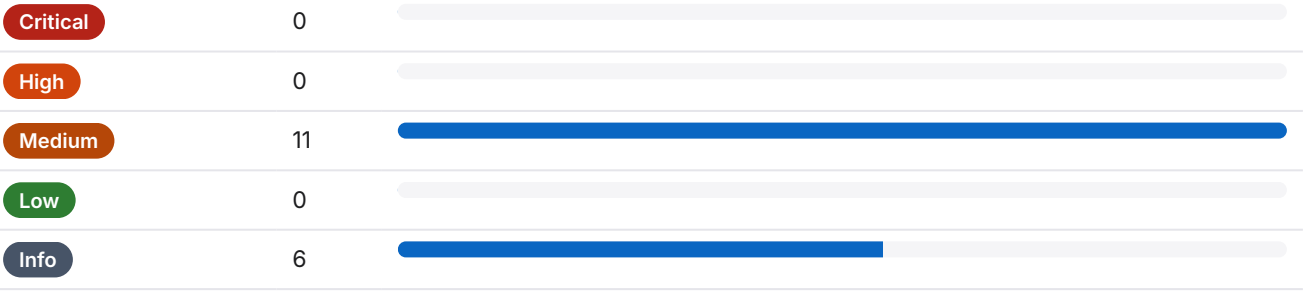
95
Indicative posture index (0–100)

66.2%
Quantum-vulnerable share of inventory

136
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-06382	Transaction signing for 3 chain(s) relies on elliptic-curve signatures	Long-term post-quantum risk	Medium	High
QSI-2026-06373	Quantum-vulnerable ECDSA: firmware and boot verification (root of trust) (3 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-06379	Quantum-vulnerable X25519: key wrapping and data encryption at rest (1 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-06375	Quantum-vulnerable X25519: channel key exchange (TLS, Noise, handshakes) (3 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-06376	Quantum-vulnerable ECDSA: transaction and message signing (50 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-06377	Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (6 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-06381	Quantum-vulnerable Ed25519: transaction and message signing (1 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-06374	Quantum-vulnerable ECDSA: role not determined (6 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-06378	Quantum-vulnerable ECDH: role not determined (1 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-06380	Quantum-vulnerable Ed25519: role not determined (1 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-06387	crypto-bigint 0.5.5 is affected by RUSTSEC-2026-0387 (no severity rating)	Confirmed technical vulnerability	Medium	Low
QSI-2026-06388	keccak 0.1.4 is affected by GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012)	Known vulnerable component	Info	Medium
QSI-2026-06389	time 0.3.44 is affected by GHSA-r6v5-fh4h-64xc (CVE-2026-25727, RUSTSEC-2026-0009)	Known vulnerable component	Info	Medium
QSI-2026-06386	block-buffer 0.10.3 is affected by GHSA-qwgh-2vcv-g2f7	Known vulnerable component	Info	Low
QSI-2026-06384	anyhow 1.0.31 is affected by RUSTSEC-2026-0190 (no severity rating)	Known vulnerable component	Info	Low

QSI-2026-06383	No post-quantum capable cryptographic library detected	Insufficient data	Info	Medium
QSI-2026-06385	bare-metal 0.2.5: informational advisory RUSTSEC-2026-0110 (unmaintained)	Improvement recommendation	Info	Low

Detailed findings

QSI-2026-06382 · CHAINMAP-QV

Medium

Transaction signing for 3 chain(s) relies on elliptic-curve signatures

Long-term post-quantum risk | Confidence: High

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** True by construction. messages/btc.proto, cardano.proto and eth.proto define the supported chains, whose signature schemes are set by consensus rather than by the wallet. The wallet cannot migrate them unilaterally, so this is a readiness map rather than a defect.

The wallet's message definitions cover Bitcoin, Cardano, Ethereum and EVM chains. These chains authorize transactions with ECDSA, Ed25519, Schnorr (BIP-340), which Shor's algorithm breaks on a cryptographically relevant quantum computer. A wallet cannot move a chain to post-quantum signatures on its own: the chain has to adopt a scheme first.

Affected asset	chain-map
Data source	messages/btc.proto
Detection method	Message definitions (.proto) matched to QSI's table of chain signature schemes.
Analyzer	chain_map 1.2.0 · 2026-10-29 05:31 UTC
Priority score	56.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Once a CRQC exists, an attacker who sees an account's public key could derive the private key and move funds; how soon keys are exposed depends on each chain (address reuse, account model).

REPRODUCTION STEPS

```
Open the .proto files named in the evidence.
```

EVIDENCE

file_excerpt · messages/btc.proto · line 4 · SHA-256 9397cf36c17bba81…

```
package shiftcrypto.bitbox02;
```

file_excerpt · messages/cardano.proto · line 4 · SHA-256 9397cf36c17bba81…

```
package shiftcrypto.bitbox02;
```

file_excerpt · messages/eth.proto · line 4 · SHA-256 9397cf36c17bba81…

```
package shiftcrypto.bitbox02;
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Chains are recognized from the names of the message definitions (.proto files and packages); the schemes are the chains' documented account signatures, not an analysis of the wallet's signing code.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them. Keep the signing layer crypto-agile per chain (algorithm identifiers in messages and storage, room for larger keys and signatures) and track each chain's post-quantum proposals, so that a new scheme can be added when a chain activates it.

RETEST PROCEDURE

Re-run when a chain activates a post-quantum signature scheme or the wallet adds one.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-06373 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: firmware and boot verification (root of trust) (3 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

7 reference(s) to ECDSA across 3 file(s), e.g. src/bootloader/bootloader.c:537. Role, judged from paths, function names and nearby code: firmware and boot verification (root of trust). These public keys are fixed in the bootloader or the device: with a cryptographically relevant quantum computer (CRQC) an attacker forges a firmware signature, and replacing the keys needs a bootloader update in the field.

Affected asset	source-code:ECDSA:root_of_trust
Data source	src/bootloader/bootloader.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-29 05:31 UTC
Priority score	43.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/bootloader/bootloader.c:537 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · src/bootloader/bootloader.c · line 537 · SHA-256 a692ec93e05c715e... · File f168d61e94931ac4...

```
  536 |         // First is sanity check
  537 |         if (pukcc_ecdsa_verify(_root_pubkeys[i], sig, random, sizeof(random), curve_p256) !=
0) {
  538 |             if (pukcc_ecdsa_verify(_root_pubkeys[i], sig, hash, sizeof(hash), curve_p256) ==
0) {
```

file_excerpt · src/bootloader/bootloader.c · line 538 · SHA-256 c8910947b0750cbb... · File f168d61e94931ac4...

```
  537 |         if (pukcc_ecdsa_verify(_root_pubkeys[i], sig, random, sizeof(random), curve_p256) !=
0) {
  538 |             if (pukcc_ecdsa_verify(_root_pubkeys[i], sig, hash, sizeof(hash), curve_p256) ==
0) {
  539 |                 valid++;
```

file_excerpt · src/bootloader/bootloader.c · line 609 · SHA-256 110ab61f5a0020b3... · File f168d61e94931ac4...

```
  608 |         // First is sanity check
  609 |         if (pukcc_ecdsa_verify(pubkey, sig, random, sizeof(random), curve_p256) != 0) {
  610 |             if (pukcc_ecdsa_verify(pubkey, sig, hash, sizeof(hash), curve_p256) == 0) {
```

file_excerpt · src/bootloader/stage0/stage1_sigcheck.c · line 165 · SHA-256 40465ea406a87913... · File 5704d66ba1ca763b...

```
  164 |         const size_t signature_offset = (size_t)key_idx * BB02_STAGE1_SIGNATURE_LEN;
  165 |         const uint8_t valid = _is_zero_u8(pukcc_ecdsa_verify(
  166 |             pubkeys[key_idx],
```

file_excerpt · src/rust/bitbox02-rust-c/src/firmware_c_api.rs · line 23 · SHA-256 0239908a79adffb8... · File 029d99f0ae93aafe...

```
   22 |     let mut hal = crate::HalImpl::new();
   23 |     util::bb02_async::block_on(bitbox02_rust::secp256k1::init(&mut hal))
   24 |         .expect("secp256k1 context initialization failed");
```

file_excerpt · src/rust/bitbox02-rust-c/src/firmware_c_api.rs · line 24 · SHA-256 d2c149057a83a51f... · File 029d99f0ae93aafe...

```
   23 |     util::bb02_async::block_on(bitbox02_rust::secp256k1::init(&mut hal))
   24 |         .expect("secp256k1 context initialization failed");
   25 |     bitbox02_rust::main_loop::main_loop(&mut hal)
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Plan a hybrid firmware signature (the classical one plus ML-DSA or SLH-DSA, both required) and the path to ship the new keys in a bootloader update.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06379 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable X25519: key wrapping and data encryption at rest (1 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

4 reference(s) to X25519 across 1 file(s), e.g. src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:15. Role, judged from paths, function names and nearby code: key wrapping and data encryption at rest. Keys or data encrypted to these public keys are stored: anyone who copies them now decrypts them once a CRQC exists (harvest now, decrypt later).

Affected asset	source-code:X25519:key_wrapping
Data source	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-29 05:31 UTC
Priority score	43.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:15 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs · line 15 · SHA-256 1c153d0b1adf9cd8… · File e15d530066a76934…

```
   14 | use sha2::{Digest, Sha256};
   15 | use x25519_dalek::{PublicKey as X25519PublicKey, StaticSecret as X25519StaticSecret};
   16 | use zeroize::{Zeroize, Zeroizing};
```

file_excerpt · src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs · line 19 · SHA-256 4b67afd4d285deef… · File e15d530066a76934…

```
18 | const AUTH_SEED_LABEL: &[u8] = b"bitboxsync-auth-ed25519-seed-v1";
19 | const WRAP_SEED_LABEL: &[u8] = b"bitboxsync-wrap-x25519-seed-v1";
20 |
```

file_excerpt · src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs · line 60 · SHA-256 f8ee14a435cd94f2… · File e15d530066a76934…

```
59 |     let wrap_secret = [REDACTED]
60 |     let wrap_public_key = X25519PublicKey::from(&wrap_secret).to_bytes();
61 |
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may already have been copied needs its own assessment.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06375 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable X25519: channel key exchange (TLS, Noise, handshakes) (3 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

6 reference(s) to X25519 across 3 file(s), e.g. src/rust/bitbox-noise/src/lib.rs:16. Role, judged from paths, function names and nearby code: channel key exchange (TLS, Noise, handshakes). Traffic protected by this key exchange can be recorded now and decrypted later; a hybrid exchange keeps the classical guarantees and adds post-quantum ones.

Affected asset	source-code:X25519:channel
Data source	src/rust/bitbox-noise/src/lib.rs

Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-29 05:31 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/rust/bitbox-noise/src/lib.rs:16 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · src/rust/bitbox-noise/src/lib.rs · line 16 · SHA-256 bec1cc899b2ee576… · File 6787c6b484f801c7…

```
   15 | pub mod testing;
   16 | mod x25519;
   17 |
```

file_excerpt · src/rust/bitbox-noise/src/lib.rs · line 19 · SHA-256 57cd8f7157563d1d… · File 6787c6b484f801c7…

```
   18 | pub use noise_xx::{Error, HandshakeHash, HandshakeResult, State};
   19 | pub use x25519::{X25519, genkey};
   20 |
```

file_excerpt · src/rust/bitbox-noise/src/noise_xx.rs · line 8 · SHA-256 784affd9139fdfbf… · File 4ade64136c560f52…

```
    7 |
    8 | use crate::x25519::{PrivateKey, PublicKey, X25519, genkey};
    9 | use noise_rust_crypto::{ChaCha20Poly1305, Sha256, sensitive::Sensitive};
```

file_excerpt · src/rust/bitbox-noise/src/x25519.rs · line 49 · SHA-256 43d70ab747c41c47… · File 63617ea3d8acc0f0…

```
   48 |     fn pubkey(k: &Self::Key) -> Self::Pubkey {
   49 |         let static_secret = [REDACTED]
   50 |         *x25519_dalek::PublicKey::from(&static_secret).as_bytes()
```

file_excerpt · src/rust/bitbox-noise/src/x25519.rs · line 54 · SHA-256 fae4ff917820e021… · File 63617ea3d8acc0f0…

```
   53 |     fn dh(k: &Self::Key, pk: &Self::Pubkey) -> Result<Self::Output, ()> {
   54 |         let k = x25519_dalek::StaticSecret::from(*k.deref());
   55 |         let pk = x25519_dalek::PublicKey::from(*pk);
```

file_excerpt · src/rust/bitbox-noise/src/x25519.rs · line 55 · SHA-256 9cb9b67a0b6ccdef… · File 63617ea3d8acc0f0…

```
   54 |         let k = x25519_dalek::StaticSecret::from(*k.deref());
   55 |         let pk = x25519_dalek::PublicKey::from(*pk);
   56 |         Ok(*k.diffie_hellman(&pk).as_bytes())
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Move the handshake to a hybrid key exchange (X25519MLKEM768 in TLS 1.3; a hybrid KEM in Noise-based protocols).

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06376 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: transaction and message signing (50 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

161 reference(s) to ECDSA across 50 file(s), e.g. src/rust/bitbox-secp256k1/build.rs:6. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:ECDSA:transaction_signing
Data source	src/rust/bitbox-secp256k1/build.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-29 05:31 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/rust/bitbox-secp256k1/build.rs:6 and inspect the quoted lines; the evidence lists further
files.
```

EVIDENCE

file_excerpt · src/rust/bitbox-secp256k1/build.rs · line 6 · SHA-256 c0e8a8076dd2e70e... · File 91983851f534bb1e...

```
    5 | fn main() {
    6 |     let secp_dir = PathBuf::from("depend/secp256k1-zkp");
    7 |     let callbacks = PathBuf::from("src/default_callbacks.c");
```

file_excerpt · src/rust/bitbox-secp256k1/build.rs · line 18 · SHA-256 25e9545e63d24ed0... · File 91983851f534bb1e...

```
   17 |     build
   18 |         .file(secp_dir.join("src/secp256k1.c"))
   19 |         .file(secp_dir.join("src/precomputed_ecmult.c"))
```

file_excerpt · src/rust/bitbox-secp256k1/build.rs · line 37 · SHA-256 47d7abfe9efe1b2b... · File 91983851f534bb1e...

```
   36 |
   37 |     build.compile("secp256k1");
   38 | }
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_parsing.c · line 11 · SHA-256 9197354c850cb53b... · File 4da2ecdc0a3b9dbb...

```
   10 |
   11 | int ecdsa_signature_parse_der_lax(const secp256k1_context* ctx, secp256k1_ecdsa_signature*
sig, const unsigned char *input, size_t inputlen) {
   12 |     size_t rpos, rlen, spos, slen;
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_parsing.c · line 19 · SHA-256 234aeaaf6b28e1d2... · File 4da2ecdc0a3b9dbb...

```
   18 |     /* Hack to initialize sig with a correctly-parsed but invalid signature. */
   19 |     secp256k1_ecdsa_signature_parse_compact(ctx, sig, tmpsig);
   20 |
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_parsing.c · line 140 · SHA-256 75ad5d0cba25ca64... · File 4da2ecdc0a3b9dbb...

```
  139 |     if (!overflow) {
  140 |         overflow = !secp256k1_ecdsa_signature_parse_compact(ctx, sig, tmpsig);
  141 |     }
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_parsing.h · line 57 · SHA-256 25af7e242d92af0d... · File 7f3e8ba009f19a5f...

```
   56 |    files (such as tests.c) without the need to set an explicit -I flag,
   57 |    which would be necessary to locate secp256k1.h. */
   58 | #ifndef SECP256K1_H
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_parsing.h · line 59 · SHA-256 4c35a81e99a7eaf5... · File 7f3e8ba009f19a5f...

```
   58 | #ifndef SECP256K1_H
   59 | #include <secp256k1.h>
   60 | #endif
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_parsing.h · line 88 · SHA-256 85efcade66268dd4... · File 7f3e8ba009f19a5f...

```
87 |     const secp256k1_context* ctx,
88 |     secp256k1_ecdsa_signature* sig,
89 |     const unsigned char *input,
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_privatekey_parsing.h · line 34 · SHA-256 163cac24f6b60e79... · File 3420812d623c0f2f...

```
33 |    files (such as tests.c) without the need to set an explicit -I flag,
34 |    which would be necessary to locate secp256k1.h. */
35 | #ifndef SECP256K1_H
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06377 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (6 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

37 reference(s) to Schnorr (BIP-340) across 6 file(s), e.g. src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig.h:82. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:Schnorr (BIP-340):transaction_signing
Data source	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig.h
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-29 05:31 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig.h:82 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig.h · line 82 · SHA-256 6aae3b20f8f8b503… · File 8185d6c6ea3234ff…

```
 81 |  */
 82 | typedef struct secp256k1_schnorrsig_extraparams {
 83 |     unsigned char magic[4];
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig.h · line 86 · SHA-256 43d5f6045ac4e6b2… · File 8185d6c6ea3234ff…

```
 85 |     void *ndata;
 86 | } secp256k1_schnorrsig_extraparams;
 87 |
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig.h · line 119 · SHA-256 1e555d327421285c… · File 8185d6c6ea3234ff…

```
118 |  */
119 | SECP256K1_API int secp256k1_schnorrsig_sign32(
120 |     const secp256k1_context *ctx,
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig_halfagg.h · line 43 · SHA-256 b3d54ae6f144846b… · File 89669a3ad2c90407…

```
 42 |  */
 43 | SECP256K1_API int secp256k1_schnorrsig_inc_aggregate(
 44 |     const secp256k1_context *ctx,
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig_halfagg.h · line 70 · SHA-256 be378be4953e3b3d… · File 89669a3ad2c90407…

```
 69 |  */
 70 | SECP256K1_API int secp256k1_schnorrsig_aggregate(
 71 |     const secp256k1_context *ctx,
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig_halfagg.h · line 94 · SHA-256 b10de912dd2b9f3e… · File 89669a3ad2c90407…

```
   93 |  */
   94 | SECP256K1_API SECP256K1_WARN_UNUSED_RESULT int secp256k1_schnorrsig_aggverify(
   95 |     const secp256k1_context *ctx,
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/musig/session_impl.h · line 621 · SHA-256 6254128e9a575e14… · File 45c89618be2ef40d…

```
  620 |     secp256k1_musig_nonce_process_internal(ctx, &session_i.fin_nonce_parity, fin_nonce,
&session_i.noncecoef, aggnonce_pts, agg_pk32, msg32);
  621 |     secp256k1_schnorrsig_challenge(secp256k1_get_hash_context(ctx), &session_i.challenge,
fin_nonce, msg32, 32, agg_pk32);
  622 |
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h · line 33 · SHA-256 fc2c5559f32fc9bd… · File 9655992c7fd21423…

```
   32 |         msg[1] = i >> 8;
   33 |         CHECK(secp256k1_schnorrsig_sign_custom(data->ctx, sig, msg, MSGLEN, data-
>keypairs[i], NULL));
   34 |     }
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h · line 44 · SHA-256 4f2e9b7b2529431d… · File 9655992c7fd21423…

```
   43 |         CHECK(secp256k1_xonly_pubkey_parse(data->ctx, &pk, data->pk[i]) == 1);
   44 |         CHECK(secp256k1_schnorrsig_verify(data->ctx, data->sigs[i], data->msgs[i], MSGLEN,
&pk));
   45 |     }
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h · line 80 · SHA-256 0ca8c9b515200145… · File 9655992c7fd21423…

```
   79 |         CHECK(secp256k1_keypair_create(data.ctx, keypair, sk));
   80 |         CHECK(secp256k1_schnorrsig_sign_custom(data.ctx, sig, msg, MSGLEN, keypair, NULL));
   81 |         CHECK(secp256k1_keypair_xonly_pub(data.ctx, &pk, NULL, keypair));
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06381 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: transaction and message signing (1 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

2 reference(s) to Ed25519 across 1 file(s), e.g. src/rust/bitbox02-rust/src/keystore/ed25519.rs:10. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:Ed25519:transaction_signing
Data source	src/rust/bitbox02-rust/src/keystore/ed25519.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-29 05:31 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/rust/bitbox02-rust/src/keystore/ed25519.rs:10 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · src/rust/bitbox02-rust/src/keystore/ed25519.rs · line 10 · SHA-256 0d1f71033a48c3c6… · File 85cb066d8f6e599c…

```
 9 | use digest::Digest;
10 | use ed25519_dalek::VerifyingKey;
11 | use ed25519_dalek::hazmat::{ExpandedSecretKey, raw_sign};
```

file_excerpt · src/rust/bitbox02-rust/src/keystore/ed25519.rs · line 11 · SHA-256 449e9e250d19db94… · File 85cb066d8f6e599c…

```
10 | use ed25519_dalek::VerifyingKey;
11 | use ed25519_dalek::hazmat::{ExpandedSecretKey, raw_sign};
12 |
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06374 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: role not determined (6 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** Role resolved for all six files, and they are three different roles rather than one. src/pukcc/pukcc.c:181 is pukcc_ecdsa_verify, the ATSAM PUKCC hardware public-key controller performing ECDSA *verification* — the bootloader's signature check, so a root-of-trust item, and verification uses public keys only, which puts it outside any timing concern. src/rust/streaming-silent-payments/{lib,hash}.rs is BIP-352 silent payments, i.e. address derivation on secp256k1. src/rust/util/src/p256.rs is the U2F/FIDO P-256 signing described in the crypto-bigint decision above. Each has a different migration deadline, which is why leaving them merged as "role not determined" would be the wrong input to a plan.

10 reference(s) to ECDSA across 6 file(s), e.g. src/pukcc/pukcc.c:181. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:ECDSA:general
Data source	src/pukcc/pukcc.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-29 05:31 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/pukcc/pukcc.c:181 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · src/pukcc/pukcc.c · line 181 · SHA-256 53085ff2be44540f… · File b5049dd725362a42…

```
180 | // Returns 0 on success
181 | uint8_t pukcc_ecdsa_verify(
182 |     const uint8_t* public_key,
```

file_excerpt · src/pukcc/pukcc.h · line 30 · SHA-256 dae3c783558d1997… · File a7650ddfd0bfe8a9…

```
 29 |
 30 | uint8_t pukcc_ecdsa_verify(
 31 |     const uint8_t* public_key,
```

file_excerpt · src/rust/bitbox02-rust-c/src/lib.rs · line 23 · SHA-256 e527a51a0f0f62d0… · File c872df632ac63a1f…

```
 22 | #[cfg(feature = "factory-setup")]
 23 | mod secp256k1;
 24 |
```

file_excerpt · src/rust/streaming-silent-payments/src/hash.rs · line 7 · SHA-256 33201f813f216ace… · File ec49885fa0902faf…

```
  6 | use bitcoin::hashes::{Hash, HashEngine, sha256t_hash_newtype};
  7 | use bitcoin::secp256k1::{PublicKey, Scalar};
  8 |
```

file_excerpt · src/rust/streaming-silent-payments/src/lib.rs · line 12 · SHA-256 08a764387fbe8ba1… · File 4e33062aad0ad482…

```
 11 | use bitcoin::hashes::Hash;
 12 | use bitcoin::secp256k1::{self, PublicKey, Scalar, Secp256k1, SecretKey, XOnlyPublicKey};
 13 |
```

file_excerpt · src/rust/streaming-silent-payments/src/lib.rs · line 43 · SHA-256 08787745f17bf569… · File 4e33062aad0ad482…

```
 42 | pub struct SilentPayment<'a> {
 43 |     secp: &'a Secp256k1<secp256k1::All>,
 44 |     network: Network,
```

file_excerpt · src/rust/streaming-silent-payments/src/lib.rs · line 124 · SHA-256 14bc8680433716b8… · File 4e33062aad0ad482…

```
123 | fn create_dleq_proof(
124 |     secp: &Secp256k1<secp256k1::All>,
125 |     a_sum: &SecretKey,
```

file_excerpt · src/rust/util/src/p256.rs · line 28 · SHA-256 925484476bdd2348… · File 7662a6309089b842…

```
 27 | ) {
 28 |     use p256::ecdsa::{Signature, SigningKey, signature::hazmat::PrehashSigner};
 29 |     let signing_key = SigningKey::from_slice(private_key.as_ref()).unwrap();
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06378 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDH: role not determined (1 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

4 reference(s) to ECDH across 1 file(s), e.g. src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench.c:24. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:ECDH:general
Data source	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-29 05:31 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench.c:24 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench.c · line 24 · SHA-256 6ec91ca104f465f4… · File 832edacf541809e3…

```
23 | #ifdef ENABLE_MODULE_ECDH
24 |     printf("    - ECDH key exchange (optional module)\n");
25 | #endif
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench.c · line 54 · SHA-256 b21ecd8d53157d4f… · File 832edacf541809e3…

```
53 | #ifdef ENABLE_MODULE_ECDH
54 |     printf("    ecdh              : ECDH key exchange algorithm\n");
55 | #endif
```

file_excerpt · src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench.c · line 68 · SHA-256 ae7392892e7a0840… · File 832edacf541809e3…

```
67 |     printf("    ellswift_keygen   : ElligatorSwift key generation\n");
68 |     printf("    ellswift_ecdh     : ECDH on ElligatorSwift keys\n");
69 | #endif
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06380 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: role not determined (1 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** Role resolved: the single anchor is src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:11, the same file that carries the X25519 at-rest row. BitBoxSync wraps synchronised data to a recipient public key, so this is key wrapping and encryption at rest, and harvest-now-decrypt-later applies: anything copied today is readable once a CRQC exists. Highest-urgency class in this repository, because unlike signing it cannot be fixed retroactively.

1 reference(s) to Ed25519 across 1 file(s), e.g. src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:11. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:Ed25519:general
Data source	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-29 05:31 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:11 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs · line 11 · SHA-256 3637a75aa4245184… · File e15d530066a76934…

```
10 | use chacha20poly1305::aead::{AeadInPlace, KeyInit};
11 | use ed25519_dalek::VerifyingKey;
12 | use hkdf::Hkdf;
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06387 · DEP-ADVISORY-UNRATED

Medium

crypto-bigint 0.5.5 is affected by RUSTSEC-2026-0387 (no severity rating)

Confirmed technical vulnerability | Confidence: Low

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** crypto-bigint 0.5.5 (multiple timing variability issues in operations intended to be constant-time) is reachable in shipped firmware, and the path is worth stating exactly because it is narrower than it looks. src/rust/util/Cargo.toml:17 takes p256 as an *optional* dependency with features ["arithmetic", "ecdsa"], and lib.rs guards it with #[cfg(feature = "p256")]. That feature is switched on in exactly one place: src/rust/bitbox02-rust-c/Cargo.toml, inside the app-u2f feature block ("util/p256"). So p256 → elliptic-curve → crypto-bigint is compiled in whenever the U2F/FIDO app is built — the Multi edition — and is absent from a Bitcoin-only build. What it does there is secret-dependent: util/src/p256.rs exports rust_p256_sign, which calls SigningKey::sign_prehash with a private key over the C FFI, i.e. U2F authentication signatures on NIST P-256. The keys at stake are therefore FIDO credentials, NOT bitcoin spending keys: those go through bitbox-secp256k1 (libsecp256k1 in C), which this advisory does not touch. Stated precisely: no exploit is demonstrated, and the attacker model is physical possession plus timing measurement, which is the threat a hardware wallet exists to resist — hence medium rather than low. Fix available in crypto-bigint 0.7.6; maintainers should match the advisory's named operations against p256's scalar arithmetic. automated severity: Low.

OSV.dev reports RUSTSEC-2026-0387 for crates.io package crypto-bigint at version 0.5.5: Multiple timing variability issues in operations intended to be constant-time. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/crypto-bigint@0.5.5
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-29 05:31 UTC
Priority score	27.7 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0387
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0387 · manifest · SHA-256 eab63427da988e95…


```
{"id": "RUSTSEC-2026-0387", "aliases": [], "modified": "2026-10-24T03:45:02.365137418Z", "summary":
"Multiple timing variability issues in operations intended to be constant-time", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.7.6) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0387

QSI-2026-06388 · DEP-KNOWN-ADVISORY

Info

keccak 0.1.4 is affected by GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012)

Known vulnerable component | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): not applicable to this code.** keccak 0.1.4's unsoundness is in the opt-in ARMv8 assembly backend behind the asm Cargo feature, which is off by default and which no manifest in src/rust enables. Its consumers are bitbox02-rust and sha3, both using the portable implementation. Upgrading to 0.1.6 is still the cheap move during routine maintenance.

OSV.dev reports GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012) for crates.io package keccak at version 0.1.4: Unsoundness in opt-in ARMv8 assembly backend for `keccak`. The affected code sits behind the `asm` Cargo feature, off by default, which no audited manifest enables: the advisory does not apply unless another dependency enables it.

Affected asset	dependency:crates.io/keccak@0.1.4
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-29 05:31 UTC
Priority score	35.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-3288-p39f-rqpv
```

EVIDENCE

tool_output · OSV.dev GHSA-3288-p39f-rqpv · manifest · SHA-256 638750a493b76f12...


```
{"id": "GHSA-3288-p39f-rqpv", "aliases": ["RUSTSEC-2026-0012"], "modified":
"2026-09-10T03:50:59.332505686Z", "summary": "Unsoundness in opt-in ARMv8 assembly backend for
`keccak`", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:L/AC:H/AT:P/PR:N/UI:N/VC:N/VI:L/
VA:N/SC:N/SI:N/SA:N/E:U"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.1.6).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-3288-p39f-rqpv

QSI-2026-06389 · DEP-KNOWN-ADVISORY

Info

time 0.3.44 is affected by GHSA-r6v5-fh4h-64xc (CVE-2026-25727, RUSTSEC-2026-0009)

Known vulnerable component | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): not applicable to this code.** time 0.3.44's stack-exhaustion issue is in the RFC 2822 and RFC 3339 parsing entry points (time::Date::parse, time::OffsetDateTime::parse and friends). time is pulled in by the workspace's own util crate, which uses it for datetime formatting in util/src/datetime.rs, not for parsing attacker-supplied strings, and no audited file names the affected functions.

OSV.dev reports GHSA-r6v5-fh4h-64xc (CVE-2026-25727, RUSTSEC-2026-0009) for crates.io package time at version 0.3.44: time vulnerable to stack exhaustion Denial of Service attack. None of the affected functions (time::Date::parse, time::OffsetDateTime::parse, time::PrimitiveDateTime::parse, time::Time::parse, time::UtcDateTime::parse, time::UtcOffset::parse ...) is named in the audited code: the advisory does not apply unless they are reached indirectly (traits, macros, other dependencies).

Affected asset	dependency:crates.io/time@0.3.44
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-29 05:31 UTC
Priority score	35.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-r6v5-fh4h-64xc
```

EVIDENCE

tool_output · OSV.dev GHSA-r6v5-fh4h-64xc · manifest · SHA-256 f9407c25f7cc6bac...


```
{"id": "GHSA-r6v5-fh4h-64xc", "aliases": ["CVE-2026-25727", "RUSTSEC-2026-0009"], "modified":
"2026-09-10T03:50:35.583241382Z", "summary": "time vulnerable to stack exhaustion Denial of Service
attack", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:N/AC:H/AT:N/PR:L/UI:A/VC:N/VI:N/
VA:H/SC:N/SI:N/SA:H"}], "declared_in_manifest": true}
```

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.3.47).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-r6v5-fh4h-64xc

QSI-2026-06386 · DEP-KNOWN-ADVISORY

Info

block-buffer 0.10.3 is affected by GHSA-qwgh-2vcv-g2f7

Known vulnerable component | Confidence: Low

> **Analyst review (QSI review, 2026-10-28): not applicable to this code.** block-buffer 0.10.3 reaches this tree through digest, so every hash in it (sha2, sha3, blake2, hmac, ripemd, the dalek curves) sits above it and the dependency really is there. But the advisory is that a panic leaves the inline buffer position corrupted, which only matters if the panic is caught and the same buffer is then reused. This is no_std firmware on Cortex-M: std::panic::catch_unwind does not exist in that build. Checked rather than assumed — of 30 catch_unwind occurrences in the repository, 26 are in external/vendor third-party crates and the 4 under src/ are all test code: a #[cfg(test)] mod in bitbox-executor/src/lib.rs, two TestingHal tests in bitbox02-rust/src/hww.rs and async_usb.rs, and the integration test bitbox-secp256k1/tests/context.rs. The precondition cannot occur on the device. automated severity: Medium.

OSV.dev reports GHSA-qwgh-2vcv-g2f7 for crates.io package block-buffer at version 0.10.3: block_buffer: panic corrupts inline buffer position. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/block-buffer@0.10.3
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-29 05:31 UTC
Priority score	26.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-qwgh-2vcv-g2f7
```

EVIDENCE

tool_output · OSV.dev GHSA-qwgh-2vcv-g2f7 · manifest · SHA-256 ac22c9ee6301d8c1...

{"id": "GHSA-qwgh-2vcv-g2f7", "aliases": [], "modified": "2026-08-19T19:30:06.686866977Z",
"summary": "block_buffer: panic corrupts inline buffer position", "severity": [{"type": "CVSS_V4",
"score": "CVSS:4.0/AV:N/AC:L/AT:P/PR:N/UI:N/VC:L/VI:L/VA:N/SC:N/SI:N/SA:N"}],
"declared_in_manifest": false}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (none published yet; see the advisory for mitigations).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-qwgh-2vcv-g2f7

QSI-2026-06384 · DEP-ADVISORY-UNRATED

Info

anyhow 1.0.31 is affected by RUSTSEC-2026-0190 (no severity rating)

Known vulnerable component | Confidence: Low

> **Analyst review (QSI review, 2026-10-28): not applicable to this code.** anyhow 1.0.31 (unsoundness in Error::downcast_mut) enters only through prost-derive, which is a proc-macro crate, so it runs in the code generator at build time and is never linked into the firmware image. Traced in src/rust/Cargo.lock: anyhow ← prost-derive ← prost ← bitbox-proto ← bitbox02-rust ← bitbox02-rust-c. Error::downcast_mut is not called anywhere in the audited tree. automated severity: Low.

OSV.dev reports RUSTSEC-2026-0190 for crates.io package anyhow at version 1.0.31: Unsoundness in `Error::downcast_mut()`. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code does not name the affected functions (anyhow::Error::downcast_mut); anyhow comes in through other dependencies, which may still call them.

Affected asset	dependency:crates.io/anyhow@1.0.31
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-29 05:31 UTC
Priority score	20.9 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0190
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0190 · manifest · SHA-256 12d4b3eeb5a7dc67…


```
{"id": "RUSTSEC-2026-0190", "aliases": [], "modified": "2026-06-29T14:15:03.650725453Z", "summary":
"Unsoundness in `Error::downcast_mut()`", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (1.0.103) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0190

QSI-2026-06383 · DEP-NO-PQC-LIBRARY

Info

No post-quantum capable cryptographic library detected

Insufficient data | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** Accurate and the single most quotable row for a migration budget: across 40 dependency manifests the tree declares curve25519-dalek, ecdsa, ed25519-dalek, p256, secp256k1 and x25519-dalek, and nothing PQC-capable. On a Cortex-M target that is not an oversight — ML-DSA and SLH-DSA signature and key sizes are the constraint, and a no_std PQ implementation that fits the flash and RAM budget is itself the engineering problem. Worth stating as a finding precisely because the fix is not "add a crate".

Across 40 dependency manifests, QSI found crypto libraries (curve25519-dalek, ecdsa, ed25519-dalek, p256, secp256k1, x25519-dalek) but none from its PQC-capable list.

Affected asset	dependencies
Data source	dependency manifests
Detection method	Dependency name matching.
Analyzer	dependencies 1.2.0 · 2026-10-29 05:31 UTC
Priority score	15.0 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Readiness indicator only: PQC support may come from the platform (OpenSSL, OS, cloud KMS) rather than application dependencies.

REPRODUCTION STEPS

```
Review the dependency manifests listed in the evidence.
```

EVIDENCE

dependency · src/rust/Cargo.lock, src/rust/Cargo.toml, src/rust/bins/bitbox03-boot0/Cargo.toml, src/rust/bins/bitbox03-boot1/Cargo.toml, src/rust/bins/bitbox03-factorysetup/Cargo.toml, src/rust/bins/bitbox03-firmware/Cargo.toml, src/rust/bitbox-aes/Cargo.toml, src/rust/bitbox-board-stm32u5a9j-dk/Cargo.toml, src/rust/bitbox-board-stm32u5a9j-dk-build/Cargo.toml, src/rust/bitbox-board-stm32u5a9j-dk-sys/Cargo.toml, src/rust/bitbox-boot-utils/Cargo.toml, src/rust/bitbox-bytequeue/Cargo.toml, src/rust/bitbox-core- · SHA-256 bf8c520f29fbbee5…

```
Crypto libraries: curve25519-dalek, ecdsa, ed25519-dalek, p256, secp256k1, x25519-dalek
```

LIMITATIONS OF THE CHECK

Absence of evidence is not evidence of absence: system libraries and transitive dependencies may not be listed.

RECOMMENDATION

Confirm where PQC support will come from (TLS terminator, KMS/HSM, library upgrade).

RETEST PROCEDURE

Re-scan after upgrades.

FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM); FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA)

QSI-2026-06385 · DEP-ADVISORY-INFORMATIONAL

Info

bare-metal 0.2.5: informational advisory RUSTSEC-2026-0110 (unmaintained)

Improvement recommendation | Confidence: Low

> **Analyst review (QSI review, 2026-10-28): not applicable to this code.** bare-metal 0.2.5 is flagged as deprecated, not vulnerable; the advisory reports no defect. Recording the decision to keep it, which is what an informational advisory asks for. It is relevant to post-quantum readiness only in the weak sense that an unmaintained crate will not gain PQ support, and this one carries no cryptography at all.

OSV.dev lists RUSTSEC-2026-0110 for crates.io package bare-metal at version 0.2.5 as informational (unmaintained): bare-metal is deprecated. It reports no vulnerability; an unmaintained package will not receive fixes for future ones.

Affected asset	dependency:crates.io/bare-metal@0.2.5
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-29 05:31 UTC
Priority score	11.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

No known vulnerability. Future security fixes may not be released.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0110
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0110 · manifest · SHA-256 1377cbc63847011c...


```
{"id": "RUSTSEC-2026-0110", "aliases": [], "modified": "2026-04-24T09:45:08.136572Z", "summary":
"bare-metal is deprecated", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time.

RECOMMENDATION

Plan a move to a maintained alternative, or record the decision to keep it.

RETEST PROCEDURE

Re-run the dependency scan after the change.

OSV advisory RUSTSEC-2026-0110

Evidence

ID	Type	Data source	SHA-256
QSI-2026-06382	file_excerpt	messages/btc.proto (line 4)	9397cf36c17bba81b4bbbc6db29f3dc450925e4566727b47ce57
QSI-2026-06382	file_excerpt	messages/cardano.proto (line 4)	9397cf36c17bba81b4bbbc6db29f3dc450925e4566727b47ce57
QSI-2026-06382	file_excerpt	messages/eth.proto (line 4)	9397cf36c17bba81b4bbbc6db29f3dc450925e4566727b47ce57
QSI-2026-06373	file_excerpt	src/bootloader/bootloader.c (line 537)	a692ec93e05c715e251b70d5df1203b876ed7d82bd28e9c6b1c1
QSI-2026-06373	file_excerpt	src/bootloader/bootloader.c (line 538)	c8910947b0750cbb6c7959f9780e84bfbe3cd04a9d2bb09cbbf7
QSI-2026-06373	file_excerpt	src/bootloader/bootloader.c (line 609)	110ab61f5a0020b335e42ad45c11bc760596590d7806f6ab45ae
QSI-2026-06373	file_excerpt	src/bootloader/stage0/stage1_sigcheck.c (line 165)	40465ea406a87913706627f03d6592256e6307cc6510acfb3fd0
QSI-2026-06373	file_excerpt	src/rust/bitbox02-rust-c/src/firmware_c_api.rs (line 23)	0239908a79adffb8db0de707bb0d8d79652f9eee0c4c289c4162
QSI-2026-06373	file_excerpt	src/rust/bitbox02-rust-c/src/firmware_c_api.rs (line 24)	d2c149057a83a51f66f258261f3e75f6e5d333f7e7cf8e4fb0e0
QSI-2026-06379	file_excerpt	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs (line 15)	1c153d0b1adf9cd8ecf03da5502f8e68db447b4ab09e5e61188f
QSI-2026-06379	file_excerpt	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs (line 19)	4b67afd4d285deeff00b1d8a4e6ef086fbb387cad74ce164645
QSI-2026-06379	file_excerpt	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs (line 60)	f8ee14a435cd94f2c3694c380ab36b8cca17c6d237875854295e
QSI-2026-06375	file_excerpt	src/rust/bitbox-noise/src/lib.rs (line 16)	bec1cc899b2ee57631342068081612e177c1e131b0062f84dbf4
QSI-2026-06375	file_excerpt	src/rust/bitbox-noise/src/lib.rs (line 19)	57cd8f7157563d1d7dcdedce2582f83213eb268d9f974d8d8276
QSI-2026-06375	file_excerpt	src/rust/bitbox-noise/src/noise_xx.rs (line 8)	784affd9139fdfbf69bb890deb430326b00ccb48a47de6340bd5
QSI-2026-06375	file_excerpt	src/rust/bitbox-noise/src/x25519.rs (line 49)	43d70ab747c41c47e17b5ba00fdf0933116113165cc5310d5e74
QSI-2026-06375	file_excerpt	src/rust/bitbox-noise/src/x25519.rs (line 54)	fae4ff917820e0219e7ac326306ae796207dce8f7af4ff5e75fc
QSI-2026-06375	file_excerpt	src/rust/bitbox-noise/src/x25519.rs (line 55)	9cb9b67a0b6ccdefe882b0cd12a8a8e62cc7886bd8c3fa9cc441
QSI-2026-06376	file_excerpt	src/rust/bitbox-secp256k1/build.rs (line 6)	c0e8a8076dd2e70e550d1aa3c11d2672989702f68a84802e0863
QSI-2026-06376	file_excerpt	src/rust/bitbox-secp256k1/build.rs (line 18)	25e9545e63d24ed02198281c8da70ca8d5106db13c770d3cb056

QSI-2026-06376	file_excerpt	src/rust/bitbox-secp256k1/build.rs (line 37)	47d7abfe9efe1b2bea40a263634d29da5de554eb838641f36186
QSI-2026-06376	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_parsing.c (line 11)	9197354c850cb53bbc475b348c0f97bb3f550706fcced8bf4833
QSI-2026-06376	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_parsing.c (line 19)	234aeaaf6b28e1d2ff43854e60a3119e3c3f55229e18841e2dc7
QSI-2026-06376	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_parsing.c (line 140)	75ad5d0cba25ca64adb6d971da937879704b4abc8845c17dc1da
QSI-2026-06376	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_parsing.h (line 57)	25af7e242d92af0d3c46ae8d8a48030e247c4bfdeba65068030e
QSI-2026-06376	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_parsing.h (line 59)	4c35a81e99a7eaf5ea7315346f3b8cacb0832f7c4a3b1a3d003f
QSI-2026-06376	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_parsing.h (line 88)	85efcade66268dd4d0b42e8f31b634ea4cd93d9d55cdc2e03671
QSI-2026-06376	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_privatekey_parsing.h (line 34)	163cac24f6b60e798317e2383fb603da718e728035262dc9759c
QSI-2026-06377	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig.h (line 82)	6aae3b20f8f8b503597acb97fe491246b6777b5a9660b8a41542
QSI-2026-06377	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig.h (line 86)	43d5f6045ac4e6b23daf11fd6c10022dca16ca76afd775c3ee89
QSI-2026-06377	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig.h (line 119)	1e555d327421285c8a81345e58876f5f2f017571adb2f2168aaf
QSI-2026-06377	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig_halfagg.h (line 43)	b3d54ae6f144846b9769cfe141d006f33bf49fa9572bafb7307c
QSI-2026-06377	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig_halfagg.h (line 70)	be378be4953e3b3d2e2b5a4ddab26874bc43e6f72392495b730c

QSI-2026-06377	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig_halfagg.h (line 94)	b10de912dd2b9f3e510c8fe909f0b6db3a25cde95edbbe0af55b
QSI-2026-06377	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/musig/session_impl.h (line 621)	6254128e9a575e14bb87c68d5f2f12d65a6084dd7de58385e4a2
QSI-2026-06377	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h (line 33)	fc2c5559f32fc9bd9a85efef59a46eaf722b1b8e2745b837696c
QSI-2026-06377	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h (line 44)	4f2e9b7b2529431d8190d606bbe0ce49659d6e5a1e0410625d93
QSI-2026-06377	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h (line 80)	0ca8c9b51520014576b4c622e33014e8b93cc7c4bd0bc826e9cc
QSI-2026-06381	file_excerpt	src/rust/bitbox02-rust/src/keystore/ed25519.rs (line 10)	0d1f71033a48c3c6e0b8363e9a41c2f5e9ee07f08735238e9856
QSI-2026-06381	file_excerpt	src/rust/bitbox02-rust/src/keystore/ed25519.rs (line 11)	449e9e250d19db94ff94019815e29a47323da4b3df8422856845
QSI-2026-06374	file_excerpt	src/pukcc/pukcc.c (line 181)	53085ff2be44540f9ac0d6f54f71ca19c4e1ac12898aef6c3ecb
QSI-2026-06374	file_excerpt	src/pukcc/pukcc.h (line 30)	dae3c783558d199701f2f4d6fa2a10c198ed29cea93653a188e1
QSI-2026-06374	file_excerpt	src/rust/bitbox02-rust-c/src/lib.rs (line 23)	e527a51a0f0f62d0ee3a3e27b910a798f61d791ba1bc143ec92b
QSI-2026-06374	file_excerpt	src/rust/streaming-silent-payments/src/hash.rs (line 7)	33201f813f216ace1faa8378e7cbee0f992739d6b30c90a1a746
QSI-2026-06374	file_excerpt	src/rust/streaming-silent-payments/src/lib.rs (line 12)	08a764387fbe8ba1e77642c172a087172ccf897e6ed0de3209ef
QSI-2026-06374	file_excerpt	src/rust/streaming-silent-payments/src/lib.rs (line 43)	08787745f17bf5694851af8a934a9791f92c751c329c4e20cd23
QSI-2026-06374	file_excerpt	src/rust/streaming-silent-payments/src/lib.rs (line 124)	14bc8680433716b8fc52ba9a85b1577a9648567735eefacba61b
QSI-2026-06374	file_excerpt	src/rust/util/src/p256.rs (line 28)	925484476bdd234888084554a30574f13c31fd235705950f76e2
QSI-2026-06378	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench.c (line 24)	6ec91ca104f465f4b861193bee0ed1f6d91c4074fb5085511ca0
QSI-2026-06378	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench.c (line 54)	b21ecd8d53157d4f920477f57713fa34b7fe6379df61daa64f27

QSI-2026-06378	file_excerpt	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench.c (line 68)	ae7392892e7a0840c3596f3442ca87927d89cc694ec91267d2a7
QSI-2026-06380	file_excerpt	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs (line 11)	3637a75aa4245184b2de8e6f4f098698530fa03b423eab7638b9
QSI-2026-06387	tool_output	OSV.dev RUSTSEC-2026-0387 (manifest)	eab63427da988e95becf061fd8169abcb9634f743aeeb7567dfa
QSI-2026-06388	tool_output	OSV.dev GHSA-3288-p39f-rqpv (manifest)	638750a493b76f126adef65e1dd53e2ef494cf4b1d4c7edcc7db
QSI-2026-06389	tool_output	OSV.dev GHSA-r6v5-fh4h-64xc (manifest)	f9407c25f7cc6bacd2695a5be9ea893203cf87df27320f0289d5
QSI-2026-06386	tool_output	OSV.dev GHSA-qwgh-2vcv-g2f7 (manifest)	ac22c9ee6301d8c1e4b09ca6951303d71527ccba2df30c96af92
QSI-2026-06384	tool_output	OSV.dev RUSTSEC-2026-0190 (manifest)	12d4b3eeb5a7dc67549bc112051813542621c4c3bb2d3ddfb353
QSI-2026-06383	dependency	src/rust/Cargo.lock, src/rust/Cargo.toml, src/rust/bins/bitbox03-boot0/Cargo.toml, src/rust/bins/bitbox03-boot1/Cargo.toml, src/rust/bins/bitbox03-factorysetup/Cargo.toml, src/rust/bins/bitbox03-firmware/Cargo.toml, src/rust/bitbox-aes/Cargo.toml, src/rust/bitbox-board-stm32u5a9j-dk/Cargo.toml, src/rust/bitbox-board-stm32u5a9j-dk-build/Cargo.toml, src/rust/bitbox-board-stm32u5a9j-dk-sys/Cargo.toml, src/rust/bitbox-boot-utils/Cargo.toml, src/rust/bitbox-bytequeue/Cargo.toml, src/rust/bitbox-core-	bf8c520f29fbbee5a6f56af359cab24476ceec5fc694f1dd615a
QSI-2026-06385	tool_output	OSV.dev RUSTSEC-2026-0110 (manifest)	1377cbc63847011c54324d486e3a7bae12ccbda6c1c9a981451c

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	10	10
Migration time y	5	5
Assumed years to CRQC z	8 (early)	8
x + y − z	7 → migration would finish too late under this scenario	7 → migration would finish too late under this scenario

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Crypto exchange, custodian or wallet”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography

- Classical only — channel key exchange (TLS, Noise, handshakes): X25519 (3 file(s))
- Classical only — key wrapping and data encryption at rest: X25519 (1 file(s))
- Classical only — firmware and boot verification (root of trust): ECDSA (3 file(s))

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
firmware and boot verification (root of trust)	ECDSA (3)	QSI-2026-06373	Plan a hybrid firmware signature (the classical one plus ML-DSA or SLH-DSA, both required) and the path to ship the new keys in a bootloader update
key wrapping and data encryption at rest	X25519 (1)	QSI-2026-06379	Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may already have been copied needs its own assessment
channel key exchange (TLS, Noise, handshakes)	X25519 (3)	QSI-2026-06375	Move the handshake to a hybrid key exchange (X25519MLKEM768 in TLS 1.3; a hybrid KEM in Noise-based protocols)

transaction and message signing	ECDSA (53), Schnorr (BIP-340) (9), Ed25519 (4)	QSI-2026-06376, QSI-2026-06377, QSI-2026-06381, QSI-2026-06382	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them
role not determined	ECDSA (6), ECDH (1), Ed25519 (1)	QSI-2026-06374, QSI-2026-06378, QSI-2026-06380	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count
weak classically	1
reduced margin (Grover), approved	13
Grover: acceptable to NIST, below ECCG/ANSSI guidance	22
n/a	10
vulnerable (Shor)	90

Detailed findings: QSI-2026-06382, QSI-2026-06373, QSI-2026-06379, QSI-2026-06375, QSI-2026-06376, QSI-2026-06377, QSI-2026-06381, QSI-2026-06374, QSI-2026-06378, QSI-2026-06380

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit
Shor's algorithm — elliptic-curve discrete logarithms Q1: quantum computation offline, classical access to the victim	breaks	ECDH (2), ECDSA (73), Ed25519 (3), Schnorr (BIP-340) (7), X25519 (5)	Elliptic-curve keys are expected to fall before RSA keys of comparable classical strength. On blockchains, coins whose public key is already on chain (reused addresses, pay-to-public-key and Taproot outputs) are exposed first; plan the move to post-quantum signatures and avoid key reuse in the meantime.

Grover's algorithm — key search and preimages Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	ChaCha20-Poly1305 (2), HKDF (1), HMAC (9), Keccak-256 (4), RIPEMD-160 (1), SHA-2 (19)	Not a break. 256-bit keys (AES-256, ChaCha20) keep a comfortable margin; AES-128 meets NIST category 1, while some authorities ask for 256-bit keys in quantum-resistant contexts (see the AES-128 entry). Prefer 256-bit keys for long-lived data; 256-bit hashes need no change for preimage resistance.
Brassard–Høyer–Tapp — collision search Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	Keccak-256 (4), RIPEMD-160 (1), SHA-2 (19)	256-bit hashes (SHA-256, SHA3-256) remain adequate for collision resistance; 160-bit hashes are already weak classically (SHA-1 collisions are practical). Use 384/512-bit hashes where NIST categories 4–5 are required.

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Remediation plan

QSI-2026-06388 · P2 · 30 days

root-cause fix

Upgrade keccak to a fixed version (0.1.6).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012).

QSI-2026-06389 · P2 · 30 days

root-cause fix

Upgrade time to a fixed version (0.3.47).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-r6v5-fh4h-64xc (CVE-2026-25727, RUSTSEC-2026-0009).

QSI-2026-06386 · P2 · 30 days

root-cause fix

Upgrade block-buffer to a fixed version (none published yet; see the advisory for mitigations).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-qwgh-2vcv-g2f7.

QSI-2026-06387 · P3 · 60 days

root-cause fix

Triage RUSTSEC-2026-0387; upgrade crypto-bigint to a fixed version (0.7.6) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0387, or the triage is recorded.

QSI-2026-06384 · P3 · 60 days

root-cause fix

Triage RUSTSEC-2026-0190; upgrade anyhow to a fixed version (1.0.103) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0190, or the triage is recorded.

QSI-2026-06382 · P3 · 90 days

short-term risk reduction

Add ECDSA, Ed25519, Schnorr (BIP-340) usages at the wallet's transaction signing to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06373 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06379 · P3 · 90 days

short-term risk reduction

Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06375 · P3 · 90 days

short-term risk reduction

Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06376 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06377 · P3 · 90 days

short-term risk reduction

Add Schnorr (BIP-340) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06381 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06374 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06378 · P3 · 90 days

short-term risk reduction

Add ECDH usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06380 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06382 · P4 · 90 days root-cause fix

Plan migration of ECDSA, Ed25519, Schnorr (BIP-340) to ? (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ? or a hybrid scheme in use.

QSI-2026-06373 · P4 · 90 days root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-06379 · P4 · 90 days root-cause fix

Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-06375 · P4 · 90 days root-cause fix

Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-06376 · P4 · 90 days root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-06377 · P4 · 90 days root-cause fix

Plan migration of Schnorr (BIP-340) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-06381 · P4 · 90 days root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-06374 · P4 · 90 days root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-06378 · P4 · 90 days root-cause fix

Plan migration of ECDH to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	

	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-06380 · P4 · 90 days

root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-06385 · P4 · 90 days

root-cause fix

Replace bare-metal with a maintained alternative or record why it stays.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	medium
Dependencies	Change window and owner approval.
Side effects	Code changes may be required.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	bare-metal replaced, or the decision recorded.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
P2	QSI-2026-06388	Upgrade keccak to a fixed version (0.1.6).	root-cause fix
P2	QSI-2026-06389	Upgrade time to a fixed version (0.3.47).	root-cause fix
P2	QSI-2026-06386	Upgrade block-buffer to a fixed version (none published yet; see the advisory for mitigations).	root-cause fix

60 days

Priority	ID	Action	Type
P3	QSI-2026-06387	Triage RUSTSEC-2026-0387; upgrade crypto-bigint to a fixed version (0.7.6) if it applies.	root-cause fix
P3	QSI-2026-06384	Triage RUSTSEC-2026-0190; upgrade anyhow to a fixed version (1.0.103) if it applies.	root-cause fix

90 days

Priority	ID	Action	Type
P3	QSI-2026-06382	Add ECDSA, Ed25519, Schnorr (BIP-340) usages at the wallet's transaction signing to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06373	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06379	Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06375	Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06376	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06377	Add Schnorr (BIP-340) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06381	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction

P3	QSI-2026-06374	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06378	Add ECDH usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06380	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P4	QSI-2026-06382	Plan migration of ECDSA, Ed25519, Schnorr (BIP-340) to ? (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06373	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06379	Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06375	Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06376	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06377	Plan migration of Schnorr (BIP-340) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06381	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06374	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06378	Plan migration of ECDH to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06380	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

P4	QSI-2026-06385	Replace bare-metal with a maintained alternative or record why it stays.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-06382	Re-run when a chain activates a post-quantum signature scheme or the wallet adds one.
QSI-2026-06373	Re-run the inventory after refactoring.
QSI-2026-06379	Re-run the inventory after refactoring.
QSI-2026-06375	Re-run the inventory after refactoring.
QSI-2026-06376	Re-run the inventory after refactoring.
QSI-2026-06377	Re-run the inventory after refactoring.
QSI-2026-06381	Re-run the inventory after refactoring.
QSI-2026-06374	Re-run the inventory after refactoring.
QSI-2026-06378	Re-run the inventory after refactoring.
QSI-2026-06380	Re-run the inventory after refactoring.
QSI-2026-06387	Re-run the dependency scan after upgrading.
QSI-2026-06388	Re-run the dependency scan after upgrading.
QSI-2026-06389	Re-run the dependency scan after upgrading.
QSI-2026-06386	Re-run the dependency scan after upgrading.
QSI-2026-06384	Re-run the dependency scan after upgrading.
QSI-2026-06383	Re-scan after upgrades.
QSI-2026-06385	Re-run the dependency scan after the change.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.1.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.2.0
inventory_document	1.1.0
solidity	1.2.0
source_code	1.3.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.2.0
OSV.dev API	v1

Sources and normative documents

Reviewed: 2026-10-13

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Appendices

A. Cryptographic inventory

Algorithm	Parameter	Quantum status	Classical status	Count	Locations	Examples
RIPEMD-160	RIPEMD-160	weak classically	legacy	1	1	src/rust/bitbox02-rust/src/hww/api/bitcoin/policies.rs:51
ChaCha20-Poly1305	ChaCha20-Poly1305	reduced margin (Grover), approved	acceptable	2	2	src/rust/bitbox-noise/src/noise_xx.rs:9; src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:549
HKDF		reduced margin (Grover), approved	acceptable	1	1	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:12
HMAC		reduced margin (Grover), approved	acceptable	8	8	src/rust/bitbox-aes/src/lib.rs:10; src/rust/bitbox-platform-host/src/securechip.rs:94; src/rust/bitbox-securechip/src/optiga.rs:175
HMAC	HMAC-SHA-256	reduced margin (Grover), approved	acceptable	1	1	src/rust/bitbox02/src/securechip/imp_fake.rs:16
SHA-2	SHA-512	reduced margin (Grover), approved	acceptable	1	1	src/rust/bitbox02-rust/src/keystore/ed25519.rs:85
Keccak-256	Keccak-256	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	4	4	src/rust/bitbox02-rust/src/hww/api/ethereum/address.rs:17; src/rust/bitbox02-rust/src/hww/api/ethereum/sighash.rs:3; src/rust/bitbox02-rust/src/hww/api/ethereum/sign_typed_msg.rs:274
SHA-2	SHA-256	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	18	18	src/rust/bitbox-core-utils/src/random.rs:28; src/rust/bitbox-core-utils/src/salt.rs:22; src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/generator/main_impl.h:27
Crypto library p256 (RustCrypto elliptic curves)	0.13.2	n/a	acceptable	2	2	src/rust/Cargo.lock; src/rust/util/Cargo.toml

Crypto library ecdsa (python-ecdsa)	0.16.9	n/a	acceptable	1	1	src/rust/Cargo.lock
Crypto library secp256k1 (secp256k1 bindings)	0.29.0	n/a	acceptable	1	1	src/rust/Cargo.lock
Crypto library x25519-dalek (ed25519-dalek)	2.0.0	n/a	acceptable	3	3	src/rust/Cargo.lock; src/rust/bitbox-noise/Cargo.toml; src/rust/bitbox02-rust/Cargo.toml
Crypto library ed25519-dalek (ed25519-dalek)	2.1.1	n/a	acceptable	2	2	src/rust/Cargo.lock; src/rust/bitbox02-rust/Cargo.toml
Crypto library curve25519-dalek (ed25519-dalek)	4.1.3	n/a	acceptable	1	1	src/rust/Cargo.lock
ECDH		vulnerable (Shor)	acceptable	2	2	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_ecdh.h; src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench.c:24
ECDSA	secp256k1	vulnerable (Shor)	acceptable	52	52	messages/btc.proto:4; messages/eth.proto:4; src/rust/bitbox-secp256k1/build.rs:6
ECDSA		vulnerable (Shor)	acceptable	20	20	src/bootloader/bootloader.c:537; src/bootloader/stage0/stage1_sigcheck.c:165; src/i2c_ecc.c
ECDSA	P-256	vulnerable (Shor)	acceptable	1	1	src/rust/util/src/p256.rs:28
Ed25519	BIP32-Ed25519	vulnerable (Shor)	acceptable	1	1	messages/cardano.proto:4
Ed25519	Ed25519	vulnerable (Shor)	acceptable	2	2	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:11; src/rust/bitbox02-rust/src/keystore/ed25519.rs:10
Schnorr (BIP-340)	secp256k1, Taproot	vulnerable (Shor)	acceptable	1	1	messages/btc.proto:4

Schnorr (BIP-340)		vulnerable (Shor)	acceptable	6	6	src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig.h:82; src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig_halfagg.h:43; src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/musig/session_impl.h:621
X25519	X25519	vulnerable (Shor)	acceptable	5	5	src/rust/bitbox-noise/src/lib.rs:16; src/rust/bitbox-noise/src/noise_xx.rs:8; src/rust/bitbox-noise/src/x25519.rs:49

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

B. JSON

qsi.report/1 · SHA-256 0f265adc21c8c990d44fbf098fcee7eebde1b04c28396bf07b7094b1e54e848b

C. Input files

File	Size	SHA-256
messages/antiklepto.proto	284	89caa2946283a469e5c783b9dd886d99728a40dd36b0e6156abe83c025ec9f0c
messages/backup.proto	1713	4616978a3a2b14c4523d1823c1a2b48373bcf731e5e35ff59f48316ada556737
messages/backup_commands.proto	724	17a8e01da16b238061f256546900a21af5b3a5bbe150b39e5d519a360e459488
messages/bitbox02_system.proto	1509	5e3ff16c8571d177180788a164d64bfb61e0e94415307a1f5f861965839743b8
messages/bitboxsync.proto	1814	8d87a05e25ba6cfec7f68a878906677ed218d64f663f6a20b24724f0d38ec54f
messages/bluetooth.proto	705	d7944887f55ee5e5196e4c62ea60564ad2a94e9d47257e49d05dbf3c4e805a0e
messages/btc.proto	9302	6f2fe1d3b1663e2617fd9587e6ec21242d02140624c797f222db4f8a26701928
messages/cardano.proto	4114	8e11c3628cc98b02d5904edf3a4d21e0492bad53ad74c89009b086e5ff111523
messages/common.proto	870	369ca12555a7810d196055abb643594d0723c7807fd46e5cca8a4a4ae8421342
messages/eth.proto	5601	3e2422076d196a16b59f743651cb1d6beecd9dae5a183bfc0c813936d49642ff
messages/hww.proto	3047	1faae60899577cf627e7c54b319b34f883aa126393d94d0cc1b84471b12cbaae
messages/keystore.proto	1894	ed5c5d677e3c017c55d71ce274246a4cd55cc8a867b259eaa9be77f204bb45a7
messages/mnemonic.proto	282	9cd94eff3b2bdd7a9d60981db9f487d2104119088ff7a912a8536654d571024b
messages/perform_attestation.proto	436	632df1278569322d391e4648de139ddf81ca83b26b5d14a0b3f0818605c0d5d5
messages/system.proto	195	706284195da36379d9d511f54e0ecc6a910b8fa8fa8158a47459567129fde382
src/atecc/atecc.c	24952	b08f8a566a7090a9f9ca736145e9565a28add3f47c8572f3a3013d56551d7266
src/atecc/atecc.h	1332	fa90590d9c07db88f66b7714dd7e74603c2f5c228332228df162c9e3e1ce3af0
src/bootloader/boot_args.h	1954	4211ed331ec60a4cebd8a66a05bc924d06a1eaf6808d6247d96c5a495b57cd7f
src/bootloader/bootloader.c	36479	f168d61e94931ac4ebd646e1317716b95b7ceb6ad14a2642b08d46ef28fe4d6e

src/bootloader/bootloader.h	589	06c21dcaacb655508bf4cf0f7fce19c7b556c589717595a45fb348c1c3b81051
src/bootloader/bootloader_format.c	2480	a8d11e946c4155c4ad0bd9cde5c67e6bd7bb2677af3e2a14dae78252e1720c43
src/bootloader/bootloader_format.h	1521	201e9cc589180cef9de3abfce4b2e98139402faf6b99ebe9accf5a8313f1b583
src/bootloader/bootloader_product.h	1455	bc0884527d99033bc323721392226a690153d3f874fd573493e02a3d1db6c467
src/bootloader/mpu_regions.c	4998	ce6c37a9c92a7ef982192a5e7e1df86a4f147fca5c9f22735276606f4124f870
src/bootloader/mpu_regions.h	633	2e0d365b60c5f4af7308cc1a3bf726beabda6581ad8d989ad111b18062b66504
src/bootloader/stage0/stage0.c	21703	e7602bdbf3060508b511e757832adbe9a9995bc6dd13477ba481d8970abdbdeb
src/bootloader/stage0/stage0_descriptor.c	648	e466f37f5e5ab14d521c3bef5ca673a0d824a5dad24bf67b0616d4f7bb9e21a6
src/bootloader/stage0/stage0_descriptor.h	1483	9a5d822def8bdb79e9505ef0453bb8214024bc4667d893f4beecad143f650d5d
src/bootloader/stage0/stage0_flash.h	7485	923840cec8b59d2a0150b15a393a36ff4ca70547602a3769a1475073ae838c82
src/bootloader/stage0/stage0_runtime.c	1000	27538c86f302b07025f7be542832db5d2843e8fa6893bc523724968ad4486e31
src/bootloader/stage0/stage0_startup.c	1872	570ef28ad3f830c70d9713b0dbe9a2323172df00e1240964459cf0541a447881
src/bootloader/stage0/stage1_sigcheck.c	9311	5704d66ba1ca763bac3bd4e998eb68c26d63ddcc74c8912267a4d5cfecd9a083
src/bootloader/stage0/stage1_sigcheck.h	333	cd78c8accb34bfcc118cdf91425314a0a79cd67d91eeb560080cbdaecbb1a602
src/bootloader/startup.c	7383	0790af69af4ff3105e7b5be5f3107d14a66043b7ca893f402be9790b49b62a67
src/bootloader_upgrade/bootloader_upgrade.h	4938	5509b50fcadf1b81604abb566a2655e91ea46525627c4bede0ce55cf00629753
src/bootloader_upgrade/firmware_installer.c	6308	fbe4a58ce6c5ee5a49b10482f7cb082c5a5ab846d7cf4dfae408f3cf86689374
src/bootloader_upgrade/firmware_installer.h	198	db2fe193d39c09380b6cec28f9190146e837048cf3c9f7ff5473ecd2ca551bad
src/bootloader_upgrade/firmware_installer_check.c	3641	2ffa4ba9da129ca84ec2eb826fa2233378ea07bedd12e6188c32a22e7889a5b8
src/bootloader_upgrade/firmware_installer_check.h	626	e6b04cd3dba6479e1d4a52f0d018c75bfa28ec4a531cc5641b5804f8a6137df7
src/bootloader_upgrade/stage1_pubkeys.c	1453	d0dbfb66799b31c124d1d47a5536b3d0ee40afb323563a7f2104668ea43d5ec7
src/bootloader_upgrade/stage1_pubkeys.h	268	6861334872aa5690cb7d9c0f6fca8b2da3c0224ccb1f2e09c2898ef50a6cc4e1
src/common_main.c	2237	6a5a78f034fdb3ff90f6b29937b3f7370b75996d6f8599ae1f28d5426580b1ad
src/common_main.h	258	00c2607686f4b979a45b494dd6465be74028d95f3758fa9cdcccd3b4b4a09011
src/compiler_util.h	156	766b232901874eff32c3045b845cdbf8b7f26bede78a78e6ea66668b04bef762

src/da14531/da14531.c	148	0e137eb5f9ed0c6017372cebe6a13c665f179c73d7ee74c5281f6ea3d5b5ccd4
src/da14531/da14531.h	757	699f504277cca1dec2db35d3a729c3d2496af56bee2d715460a9cf8bca9007aa
src/da14531/da14531_binary.h	615	ab36b4d19ab2e92690987e13a9bb764070dc4fe1b37f582b13dfb1920dbd7378
src/da14531/da14531_handler.c	10934	470474701180c1c3a831ca43c73b3323456cc31f063eac2cc536b77fe5cd8a9c
src/da14531/da14531_handler.h	488	c4f4950fa1e7095bf80defffee2d29224512c369d4d42a51c3b37b7f7a9a0075
src/da14531/da14531_protocol.c	14092	fccf1437f23c480e2b2675366367ab4ce8d4f41ebef6584a5c729f14c0a32697
src/da14531/da14531_protocol.h	2026	b35f9a6a828becdd2ae5b9e93f8615982afd923518df14ee269d34d27493c248
src/delay.c	2214	d9b814da8a0b29b9316c7d4d9062aad566376b3f580111e44e62b572c218aaa7
src/delay.h	527	b181e4281855a1a7ee890769b72d7884157729456e11760b93319ea8065c54ff
src/factorysetup.c	40551	f23340c080fa71056e24109f64c2404ff90acf2b6bcc516e5daa267731c94e18
src/firmware.c	1346	e8c3cd856d1329684a6519d907dc516953a5da6260bb895cb8288986fa80058f
src/flags.h	4465	20b4d9f439ebac1783b6a6a9b58d3f44f5795a4da84148d9f7772c2d7234fa77
src/hardfault.c	1227	5d2a769cf3e77da97afcea5b416c9853b2c8db0db8bfb6c52a9de6345d16eea4
src/hardfault.h	858	5944df9738bce65d30ad8a83310a4ba489613bc075efd653c2807edd20232a12
src/hww.c	8090	a01aa7f5b50ae9766c9316cdfc8a4fea04f22202b5dfeec770bbb2042e3496f7
src/hww.h	1190	3dc95f986eab99ef6868410b64498f5af12e661ab68871c0c8c6f61912b85fd1
src/i2c_ecc.c	2276	ed06ab53c14f7911372708f60f040238decc514e7163c2decb99e8db658ffc96
src/i2c_ecc.h	1462	c36dd2e4b88288c1d4d5422884b2eaf273edc3973dc880ea06c30f3035a59496
src/memory/bitbox02_smarteeprom.c	4544	2d4869e79d0baa5614a11659bea610b382cf25f17fb5ba76683cacba93dc0eea
src/memory/bitbox02_smarteeprom.h	891	a014fb1fc58e11fcd563a9281dd3b992c0aa75385b54be6124ae950e85c32ad3
src/memory/memory.c	35229	10d47933c8921fb95a5c15b3e91291f273a5ace9f1ee9c98568bffc8c894820a
src/memory/memory.h	10717	f5f44c8b0780a106f193a012d03db97abf90230eed6b88491df8146610f95963
src/memory/memory_shared.c	8175	4b5997eff681a1063e56e689a03adae0a6edfa5ccddd5eb12f966c759096bb9f
src/memory/memory_shared.h	5977	b7c85bf18214d96b882c4e5ababf4bfe5194a8d29e3db15d9d07dbde50912f5b
src/memory/memory_spi.c	2980	5c7ec0943599eb710ab7d2b1681fa72fa2e7148ad26f1f2d73e90127682438e1
src/memory/memory_spi.h	1279	8810697f34c32b38045a4e398daf86d59353a28cb0b35a9bfe0bb39e4bf3b49a
src/memory/mpu.c	1183	428c3a5c05260bea2d6c8fd7a9648904c05ef48e756ce13d54ec8b2cc49a8919
src/memory/mpu.h	3189	e3e8eeaa6c7f657e82c8afbf80ee43af9268ac058bfe93c5bbe289012097de4c
src/memory/nvmctrl.c	383	51c13e773d9816fa9b3d8a4fb402ca07aec6d745072f4616c3c04a7dec09ad50
src/memory/nvmctrl.h	312	eb730b09fd1ba0120da5f956d8dfb4d2a39f5ca55f8660d8777dd1ab59717937
src/memory/smarteeprom.c	5033	28653e4583b809e72e8bb6104a23db25c6fa963dd12f0a4b7c9b142fa4d961e4
src/memory/smarteeprom.h	3092	eac71c154b3af713597f18c878aa2c2d2b72b43a811df1220b46710b27f07a42
src/memory/spi_mem.c	11033	9f2bba7727e4ee26a6e3c29b3108259feb98bce5515d8397ff9dff907ba179d8

src/memory/spi_mem.h	3912	032202c6f683a3a269ceba9e51ad78d9b11bd12924eb29b765caf3ea09fc62c2
src/optiga/optiga.c	36862	d81c6f4b45d9923458bf7b218170acfaf729a6d36876d9bea34b848e03c6a54f
src/optiga/optiga.h	4906	d992cdf0cda34979f1a75dbca2c1307f5aaf6fded168246fe36a9e771bdcad96
src/optiga/optiga_ops.c	5215	faeb33491aae26ecb601778ffad183100f4fd698bda52e9b1321d353b69b58ad
src/optiga/optiga_ops.h	1694	ab2ed1d32962652a76c14fc95fd58b1ad24d273d3a692a896d2cd23ab4a8055a
src/optiga/pal/pal.c	1558	3769a1635f88870578cbd926257df76749f1b9774f78bcf82569fab8efe527a7
src/optiga/pal/pal_gpio.c	1979	89e0f438ddc439ed69d259aa4f760fe92f41f2f5751fcd1b570b21a31d00912e
src/optiga/pal/pal_i2c.c	11391	8e9c342201674cdb6241e0b3fd9dfc7efa17100944b49c8660fff0dec82071fa
src/optiga/pal/pal_ifx_i2c_config.c	2906	e4ff9782e587b6fb265086c5dace6a2ae8d2d15709d7a79dcd085ca48e731303
src/optiga/pal/pal_logger.c	2858	ed4810ab3dbb786ec7cd1902053f342869822b9ff765497cd256ff1887a73de3
src/optiga/pal/pal_os_datastore.c	2763	178127d32e56421ee047b0cc0e6ee0afa935fa85c717c7da6cdf1c8fa2306195
src/optiga/pal/pal_os_event.c	3355	245c5a1d4c696fdbb90764e6b2b5b8dd59fd531929e856f3a57b2916eadf6ae0
src/optiga/pal/pal_os_lock.c	2603	71ffab63bbf2f395b09df0a1b198a6d418473d35b06263f64639816eefb08709
src/optiga/pal/pal_os_memory.c	470	7b29109eb25ffa9f501f444ac0e7675a8e9cea8068a395a6b85ef59864105eff
src/optiga/pal/pal_os_timer.c	2550	ecf6f183c7cb7b6c865f9e58ad392ebf6a69718c7faddceeceed0beeb8642b55
src/pac_ext.h	654	1b829db5cbc08b8d68626be28855dc5508bd39233aadb20cd5b347b39b84c0b3
src/platform/bitbox02_pins.h	1308	7b73cfee547ae31cadde79c3c727001b7f8268b132a446d7884bb78401ba9d70
src/platform/driver_init.c	13012	ce0ab53cd6fab718023fca835d52705f61e6b1b412d85fba33e74dec346e7189
src/platform/driver_init.h	2021	057862dc09feac0d0d67e191dd027d26cc0d91677b29838111b46140d1b1f745
src/platform/platform_config.h	1732	c6a8ff515fcf1c87f79c366b42870d3ad34b09adac5b02a4d6451b658b2b6374
src/platform/platform_init.c	2465	cc2d88ab9b7eee346288271974f1200f40ce6b33f233877990d1889a93961a6c
src/platform/platform_init.h	359	d4c05fefa903b6446086660da71863f44bc3baa11b76f7148caf14815c026123
src/platform/sbrk.c	1187	5552fa9172238701b473d3b0bfa0df1b1906e85037596958d12305850e73c8ad
src/pukcc/curve_p256.c	4832	f4aba16f4398bbf89102e62b826cefc8c02d3a4731a73387393d48a864a8b212
src/pukcc/curve_p256.h	211	c620f92a051fc37d6c4b976a72cadc61f24595dbfa1dfed6326dddcde19f1225
src/pukcc/pukcc.c	9798	b5049dd725362a4217e5797928ba8ad32e3c7235d029f366a411e38631151540
src/pukcc/pukcc.h	1169	a7650ddfd0bfe8a9c23dce685d1d648c412ea08b65c28f1403569071e88d5e74
src/qtouch/qtouch.c	28561	a3ce935421670987dcc50cbb7fa710d2e507fd5f5dff108a1c53ecbc3205da6c
src/qtouch/qtouch.h	10273	ece235d4e91d9b3b459490b835463f2c1e9a621ea7e327c42cc5c4eb0a917a86
src/random.c	2290	8e70d7c104bf78ace2348c526b5e4e12625ad8bb56b0938be2bcf88421054019
src/random.h	711	8c2bca629ef2ac0214138b7c3756b271540426323fb18c0a80b1d1a14ecfd64b
src/reset.c	648	817e226f032c6a0fc76fde0f4906ff0c6151f609bb4d6c38d4fc86bed4e074d0
src/reset.h	243	7050c993ba457968af7b3411d33e22754620c9aee2a8000305488e92ea3653fa

src/rust/.cargo/config.toml	1932	a6bddaebf210d9245b6e206167b95b578e86e14a44e8ebf6c588fd7ae3669b72	
src/rust/Cargo.lock	37444	26c6fdf274b15f25b33b594f5adaae1be5d54998c1ab4584a91a53be681c323a	
src/rust/Cargo.toml	4099	73bfb23592887b4a84691a76f5034f4c29830dbcfeff26eb17d8b6e5d9dbbce4	
src/rust/bins/bitbox03-boot0/Cargo.toml	602	b45eb014db8ef2e7ac79a791989422bcfbae6bf8e6bd6254b37d887b5fae035e	
src/rust/bins/bitbox03-boot0/build.rs	1188	ce807537a25ae8a54e4d07708b44fa86de7cd518bd820c4fd493cf05378d58ae	
src/rust/bins/bitbox03-boot0/src/main.rs	1411	65d8b40c8876936d6b38e3fec8aa11fdc622c8a984a0ba697abe7bdc3f01d7c2	
src/rust/bins/bitbox03-boot1/Cargo.toml	602	dd41d2262a5b31bb5233966eb1a761d326f1ceff70144a451f406eb427bcabaf	
src/rust/bins/bitbox03-boot1/build.rs	2885	56d2bc01330a026f7ecca4fbdd782a6ab33a2f99e50a75cbae2f8f449b962151	
src/rust/bins/bitbox03-boot1/src/main.rs	1452	5147e04dc133dc02a987c6d5fea26176da929eb8adbb89d36cd5786ad81a90e7	
src/rust/bins/bitbox03-factorysetup/Cargo.toml	900	7e0ad6e8bdd8aae31eac633c9da465cbc325a4b27f587e3a6f320c75ffddc1de	
src/rust/bins/bitbox03-factorysetup/build.rs	2285	d29edafa35af5298df345ada8e61b938812591fb73cf0655bb022b7375a20d9a	
src/rust/bins/bitbox03-factorysetup/src/main.rs	3628	a253e1c4415b9ffd333355e274b1ec08c9cfa1adc0eb9af9cbb97d6cfafd589d	
src/rust/bins/bitbox03-firmware/Cargo.toml	819	93fc8dab006fc6c5b9b5336b7ac129445d9a47a4aca353989674a928a3244ed1	
src/rust/bins/bitbox03-firmware/build.rs	3198	a7109b841bc352e2af96fd651504faa5bb14d263843df3f99a3e177fe2e5d179	
src/rust/bins/bitbox03-firmware/src/main.rs	556	e02dc06cc0918423fbc2fca5b373064fd2086c68ee174e9e7b41fb218f8e9860	
src/rust/bitbox-aes/Cargo.toml	315	f8c0eda103f0f2a76ea59a2ee7469215f09951ef490a28b9bb1a1c478ea0dc1c	
src/rust/bitbox-aes/src/lib.rs	189665	d1b6231d59cddb178fbde87ab0f9ab35fb8bdbdc5c1d197e8298dfb13f5f2394	
src/rust/bitbox-board-stm32u5a9j-dk/Cargo.toml	497	b41260fdc41c8a6146e9325384c72c19e3eea5bf026ca13af9e370bb0f85e87c	
src/rust/bitbox-board-stm32u5a9j-dk/build.rs	2484	342a59fec95ff796491f0382609d4885c17a70bd9c4ca4350cb5b7065cdd95c1	
src/rust/bitbox-board-stm32u5a9j-dk/src/lib.rs	629	98780551b6e10d5bdb56bca266d7c84c7bd5b2f1ec9cfb4d985cab2383af6d04	
src/rust/bitbox-board-stm32u5a9j-dk/src/memory.rs	4370	59076a19e7152ab2f5a5aab0d9005a31fdfa297bb68e03f37a18b32f32f22999	
src/rust/bitbox-board-stm32u5a9j-dk-build/Cargo.toml	331	f3b43b17755cd35be3bb7cdc2a5a3f635864ba7a72098d51c0898bdeb1608477	
src/rust/bitbox-board-stm32u5a9j-dk-build/src/lib.rs	2349	ef079b608145f4834765b3aea1cee1f7ab519853e1e9892f712d8b3fb2961721	

src/rust/bitbox-board-stm32u5a9j-dk-sys/Cargo.toml	363	e86eece21be38507a20b4c38642399bf2e5d404343c0ad8ac8cf39708cc7644c	
src/rust/bitbox-board-stm32u5a9j-dk-sys/build.rs	4573	94c13a1fdaf427adcdff8c386e2a44644c58c21ea8e7f2e7261fe8024b867922	
src/rust/bitbox-board-stm32u5a9j-dk-sys/src/lib.rs	272	1b57158882a13471de4be80cc918e22e9628a33222e9d20fdd21b2cdedeb5d45	
src/rust/bitbox-board-stm32u5a9j-dk-sys/wrapper.h	141	100e3781691854b92ceb38c0eb2f61f40f7da3c1245e7b158a6480dc670c0a5f	
src/rust/bitbox-boot-utils/Cargo.toml	143	25690b5da7d4452621cd5ab17a91bea7ffe634075f88dc5a97bba49181347b6a	
src/rust/bitbox-boot-utils/src/image_header.rs	3066	0a54aa9230fd76555626b278bfac1ea7a7a280d93ac20635a1efb26c11da3cf7	
src/rust/bitbox-boot-utils/src/lib.rs	22385	517532d8b9006f867adb50754bcdbf773e43afb4fc7e90abfbbf0ace89ef9234	
src/rust/bitbox-bytequeue/Cargo.toml	230	1b9256563d1f7b139b57c44e71673f17b4e6adf34023ebfddb15a93c839c0f7f	
src/rust/bitbox-bytequeue/src/lib.rs	11527	a297010473f7b7f5043a2cead7181cccbc5dd5acf852059bd2ccaa55f7bb00ad	
src/rust/bitbox-core-utils/Cargo.toml	535	ea9a88c066735da317bb3c937e841a12c93b005dc41cbff7647080dbefbfc1e0	
src/rust/bitbox-core-utils/src/lib.rs	103	0096bd43bcfd4ce09e49a2e95943e5bf27f8f73611d08b61dcbd9f061f5ba220	
src/rust/bitbox-core-utils/src/random.rs	4073	27fc4f4a389b1010b5ce0f37de67f00751724b1a9fdf83d10e2e601ab42a8b35	
src/rust/bitbox-core-utils/src/salt.rs	1973	0a5be445ef7f2d5a2b1df43462fff5b4dd52b7a2e5cb6e5492ca6054f61cd035	
src/rust/bitbox-da14531/Cargo.toml	401	dc7a28ec383a3eb83ecb0f364d10b1da99b32f8be098e990488693c4fa075e79	
src/rust/bitbox-da14531/src/lib.rs	7209	ff8cb7cd0e1ee5290e86ce22f3629aec737ebde2ca1ca5cf87334d3886e717a4	
src/rust/bitbox-debug/Cargo.toml	275	82ccb7a869fb1c4fcfa222bd5fa9e74389a268f45d746282a625bf40f4219a0a	
src/rust/bitbox-debug/src/lib.rs	949	140a69f4d3b87b65e651e7af56f409c94b8db6324409636938c2444879bbf32c	
src/rust/bitbox-executor/Cargo.toml	267	98befdb3a6d49e8b576b0ac2f21918db7f94b0d24620878be6e3c2674b81512a	
src/rust/bitbox-executor/src/lib.rs	7140	facd59247f5511a85d021b99e248223d4c98ca45e97a89fd2824f900fe3b061a	
src/rust/bitbox-framed-serial-link/Cargo.toml	360	0b715df72f2c74113151e0b699e79c6661e8f2091e0bb80434aab754a2ee4e3f	
src/rust/bitbox-framed-serial-link/src/lib.rs	17048	c97d73bd553608be30463d109f86b23b0284e973c5ffdaadef961aadf00dcb31	
src/rust/bitbox-hal/Cargo.toml	252	8e639839c006a38521fe6ef232ce70588f8b88a404f68bf0053526fad2d6dd6d	
src/rust/bitbox-hal/src/eeprom.rs	1948	200715d25813eeb9d50c237a7c8d4a0861cf7de64d38f81b6fe0cff849389a93	
src/rust/bitbox-hal/src/lib.rs	1913	456f9367a84f57322c65ae911c5e1aeb41c33c020343204d4b78db1d02a42e97	
src/rust/bitbox-hal/src/memory.rs	4151	ed950634d4ebb11fa50c62f323f2dcd92901986820d8c0d769146e9f31f51d4e	
src/rust/bitbox-hal/src/random.rs	172	3d53abdd13fc8d9c29b0534d41864d1f8fb97c9a04c38da2ad112fc391761b8a	
src/rust/bitbox-hal/src/sd.rs	588	7a3c3a204b2985238560a1951307ae884f86f5c9c76e66391b0ad84981381306	
src/rust/bitbox-hal/src/securechip.rs	4226	da71595a0cca42f468cb939ff2a6fbd9c62eed395caf4d4d16425a1f8d3bb36e	

src/rust/bitbox-hal/src/system.rs	1019	6da346170863b3ba80e5d7ab3e456fa322d94d0a36ab1e1a1769ceb3cfd87771
src/rust/bitbox-hal/src/timer.rs	159	ab7bfc9e02e33d3b86168f606ced6332ea33d39b5503fae655b11cae2a52dfd9
src/rust/bitbox-hal/src/ui.rs	8042	bc9fb5c0d36d1a1e879cd4d02e116998272c37f1e537d48eb6a505d20946f1b7
src/rust/bitbox-lvgl/Cargo.toml	268	daefc6d741706e60098419b9ec4d5b2ceb5a3de23878b9bc7206085aaa9cf3bb
src/rust/bitbox-lvgl/src/color.rs	473	71d7bea77d1589672a954ef696f73db64bb36192eae7d0f37898c13d04bfe829
src/rust/bitbox-lvgl/src/display.rs	5374	916b3d2e5a95aea091b5f9368115cc017ce770ed6c7f2e1fd2f001651822469e
src/rust/bitbox-lvgl/src/font.rs	1900	5f87b8d677c62d374ffe1c4d4c557abcec777306bcf8e8c877840fe449ea9d8c
src/rust/bitbox-lvgl/src/indev.rs	2037	bbf58a031fdbf5b41ef0f9b6616b54a24891ed27481c92a9b41d6d59f24bccfd
src/rust/bitbox-lvgl/src/lib.rs	3486	7b69d1adbd4abb46f93f8ab9312bf857383eb1f18acf73e3116216070dbfa9de
src/rust/bitbox-lvgl/src/log.rs	985	fe230bbebb5892fa4ea4fa3d6f3fe7b0034bf21fd318a2265db03267b814dc7d
src/rust/bitbox-lvgl/src/style.rs	3816	6db64a1617b4f2647ec3d2c6de756f7a027b773a27bd7d92f2778779ce8b30b4
src/rust/bitbox-lvgl/src/system.rs	105	ffd20d867f7d4328e68d59f135e9134ca691dd5851cd271a609b6c96b6e02036
src/rust/bitbox-lvgl/src/tick.rs	157	ad20354b120701fbd19ef4be25425cdc219eeb71e5951ea0c0ab28de87a8ee89
src/rust/bitbox-lvgl/src/timer.rs	130	3eabc6b60fd7223fbef597a37088779ffda143d1bc6985f3918f280ce5c78e3b
src/rust/bitbox-lvgl/src/util.rs	1312	f3f9f97063188336dd790b1aba5c0b085020096af6a57711c0401ae575eae491
src/rust/bitbox-lvgl/src/widgets/arc.rs	3409	0a116c3cc38d71e7fb015b929fe518a3dcbd44e7bf0d046390c0fb67f7fc2238
src/rust/bitbox-lvgl/src/widgets/bar.rs	2207	1684518d5e6da6338ef0df80538b98941221ecb55002d737b76a4eb2e1532a07
src/rust/bitbox-lvgl/src/widgets/button.rs	521	3f8a2ac68192d05fc1539694ab44e2f6f8b28c6fc72bb1d2b757474ddde8083c
src/rust/bitbox-lvgl/src/widgets/buttonmatrix.rs	6488	5024d703f760384f238652d387ded9200aa1baebce14e48d5c0848e441538121
src/rust/bitbox-lvgl/src/widgets/canvas.rs	4115	a5b70aa02b2edb1924c0945b781853a9172f0ba1f18a66dc830d2eb838896c62
src/rust/bitbox-lvgl/src/widgets/class.rs	2361	efa9537454a60a2d4db805c8139483aee33a8fd71e7a1a9d3dc59d035a8310f8
src/rust/bitbox-lvgl/src/widgets/image.rs	3280	fd27b5b532c89fea4ad00cd82e0aad8551b4b8720565d3e09a0a32047a640faa
src/rust/bitbox-lvgl/src/widgets/keyboard.rs	4508	267840305dfce7dd03cee5bbd73c66b2c0acf6484fbd6937ab9719baaf7e5ea7
src/rust/bitbox-lvgl/src/widgets/label.rs	3497	9c01936e4c2ffea788935298186fa3165630ec5d5d7993cb1aad1ef9a3220cd6
src/rust/bitbox-lvgl/src/widgets/line.rs	2822	65f099ad38111fdbd0092cea34299168ef3886c47f10a94569fd887db9a28f84
src/rust/bitbox-lvgl/src/widgets/mod.rs	2567	4a1f2a556ae514d9a87b742134f675d8f8fa041a1a832699c434a5419425e335
src/rust/bitbox-lvgl/src/widgets/obj.rs	26478	986a3fb1a9bd5628a7428f9feb7f6d9c2b0aa2436ac7e25ca5fbba866f0ba007
src/rust/bitbox-lvgl/src/widgets/slider.rs	2700	7ff614b820dbb41728c77822445f8d9acabf0eb2405fdbc54c13ca5a10bfcf2c

src/rust/bitbox-lvgl/src/widgets/span.rs	18294	d53070663d6ebff5243cbffc2a73460492fd4d5e6465a7a09947974fbe263f6e	
src/rust/bitbox-lvgl/src/widgets/spinner.rs	774	c3f69a2baf9c498ba1574b2133fb1a10fc3f66e01ca0efe6735f214d7f70b896	
src/rust/bitbox-lvgl/src/widgets/textarea.rs	6641	c0a550e929d8ef527d767c582394858c4d0a3a9eeb1b3182a4298ab68ebc2658	
src/rust/bitbox-lvgl/src/widgets/util.rs	3599	64c737a005a0aba480725702fe5539a2fbfd92ad419678a82bfd03ef3219e715	
src/rust/bitbox-lvgl-sys/Cargo.toml	248	389cb4c1c28f0ccc4a422025b75061db160cc8fdbfb68b51f77fd00e1bba91be	
src/rust/bitbox-lvgl-sys/build.rs	9866	d28e575ad9baa893bce904d3b2b74b2db3fb600a1b4ee36f6a911c7040bf7366	
src/rust/bitbox-lvgl-sys/lv_conf.h	51220	5a13d194e9d7013099caf89cea5091b78ff16a22b9c9ea928f7276c1f35bf7d1	
src/rust/bitbox-lvgl-sys/src/lib.rs	341	fcab8b13940551094a4a7a4fe477bc6186711f50e0dfa2026dfab7013858f313	
src/rust/bitbox-lvgl-sys/wrapper.h	338	acd8d58b7615012977b0147392b8c616a71205c3b2cbc66b9953a5d0bfbe1f09	
src/rust/bitbox-mcu-stm32u5/Cargo.toml	356	4980c20d6863a9faddd52520b0ea0f39205f5ea2baa2441d203e6b6cb38c08e4	
src/rust/bitbox-mcu-stm32u5/src/lib.rs	87	33f59b81e49b867519f3e09635f27a1a46f6bc4acd9d0f4aacbbc0adaca3d127	
src/rust/bitbox-noise/Cargo.toml	620	1903f4c84f747fe0de9b1af349aa437b9ca54624f36075226df2314080c4e751	
src/rust/bitbox-noise/src/lib.rs	556	6787c6b484f801c7d4e95443f0512b7247bff8dba8f871c54b92f21685e2d930	
src/rust/bitbox-noise/src/noise_xx.rs	10043	4ade64136c560f523c56ae0e567721938ae097ab6a192c547f5891d45764c599	
src/rust/bitbox-noise/src/x25519.rs	2023	63617ea3d8acc0f018c1664ce5abae381d71c29cec4b9f994bd042e92d982315	
src/rust/bitbox-platform-host/Cargo.toml	534	e13d76508db7d5e832dbcea9d159fd868d9705976d3f96491af589acac2cc51d	
src/rust/bitbox-platform-host/src/eeprom.rs	1808	e90f8286ddc5068a56e3b59f033f32e0292148117c35f826df0ee2fd36e3c259	
src/rust/bitbox-platform-host/src/lib.rs	199	aa279b040b8a9357dd020dba45c6996af8780a46caa027bd1adb80c7dbfb116f	
src/rust/bitbox-platform-host/src/memory.rs	12192	702d8734b4db0d0a259d6bc831b273411179bf57fdad065bd47c6cd677d60ed4	
src/rust/bitbox-platform-host/src/random.rs	2083	fde390232194f51b1530025eec812bda2d16781b2ff11b7fa72b68d5a472a393	
src/rust/bitbox-platform-host/src/sd.rs	3060	3f40d002d85cd970f8c70473f32197c54313c5766203d5fce33f2a5371c82b3b	
src/rust/bitbox-platform-host/src/securechip.rs	7297	a383fe084917e785e7794e8950397b3fb5d39628ea0a926efdcb1bf9746c0bad	
src/rust/bitbox-platform-host/src/timer.rs	1617	0e643681b0e7089742a91cb2621e39fa2123f634f50e908f6140751f8fa44f5e	
src/rust/bitbox-platform-stm32u5/Cargo.toml	718	5501510cdaf7eb902f231f719b19f3fec7a542b4dc3160310cacc4a5e129d3cf	
src/rust/bitbox-platform-stm32u5/build.rs	597	ab8ee6746d73105d4ea8cdb73205d7b3422b288c16cd5922c2ea5ca146b0697d	

src/rust/bitbox-platform-stm32u5/src/flash.rs	3007	a128bfb727a5d77c521eadc35601cc385be833f8d0ce7089aa9bbdddd4a59bdb
src/rust/bitbox-platform-stm32u5/src/inner.rs	2010	f3c9459d7051aab14209a7e53d6995a9c9c2aa500ff9b1c910f522f74b57def9
src/rust/bitbox-platform-stm32u5/src/lib.rs	185	2ee9aac744280d9c128b6d74eab47a33a60e8f4524aaf75cf6ae1a8c1a0041b9
src/rust/bitbox-platform-stm32u5-sys/Cargo.toml	345	f00953d4fdd7b0a339a4275ba3a016813153790f1440bda76edffc6bb65a7700
src/rust/bitbox-platform-stm32u5-sys/build.rs	4624	e5a03d01598e2c749dc94ca18a3d1c069e1726108c863a167648be5c5f450f98
src/rust/bitbox-platform-stm32u5-sys/src/lib.rs	1804	7779a494a7b1f81a99cfe6da8e827ffaafa632b1e15d543bc8b55ba93aa53ace
src/rust/bitbox-platform-stm32u5-sys/wrapper.h	67	7bdf790812e8172badd61a9db85127dff2425a4654ecad202c1840d37e733db4
src/rust/bitbox-proto/Cargo.toml	529	21decdfa25e99c6645d4bfa993c898bdace001c059e9f7fa7b0023e22a0768ec
src/rust/bitbox-proto/build.rs	808	6d4f6136eadec490c81d39aae758968650da4c2442022749a034d968f027d141
src/rust/bitbox-proto/src/generated/shiftcrypto.bitbox02.backups.rs	3421	45f3b31bcbddbb90b7b8b29ebb0fa4cde683b573ea309bfa41e9c08f4a7122af
src/rust/bitbox-proto/src/generated/shiftcrypto.bitbox02.rs	86465	5b538703f43ef6f30896a79c051a3e37caf7b6c365b5493db6d535adba0c6fad
src/rust/bitbox-proto/src/lib.rs	1372	1d36872343cc062e807a7315fead89a405251f59fb66be5980de93594fdbf647
src/rust/bitbox-secp256k1/Cargo.toml	484	77442775d85b6ad99b047180d03c077b5427273e767a06175c5befb7ffb275a6
src/rust/bitbox-secp256k1/build.rs	1565	91983851f534bb1e29d57360162a303f9847ba6378c5873aa4088c655c748f7e
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_parsing.c	3595	4da2ecdc0a3b9dbb9029d188bb0243764a3c2f2324613a688f1ac4d3fc878851
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_parsing.h	4230	7f3e8ba009f19a5f2759c9065af02cc8e62351100d69f8ae7e762fa4b65ea675
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_privatekey_parsing.c	5112	48352c75520eecd141936c0011601925510cdf60afd3e46e79f0a0547cbc78da
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/contrib/lax_der_privatekey_parsing.h	4016	3420812d623c0f2f86debe0535f98eab9b929036bb2fee2de8e3b1adc3e1a0b9
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1.h	44243	c04f49681ef8c9c5cf28ce0f9cc528ef39d64df0bace093bc3a18de4855b516e
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_bppp.h	2761	bbd36b1d30ba8d181a84a234a29b91e1af6a6610cb24b7bc61326d12279d0feb

src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_ecdh.h	2560	ad314d14cb79a79d4113862f21e0deccf37962db7811c801dab2899f91dd8222	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_ecdsa_adaptor.h	8168	c0f418ba9344aa4314ef9358fca624008db57ad17191f448b967daa16b45804e	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_ecdsa_s2c.h	12269	ff02de0c7f482bd1bc68c725293d555e265809656b26a9d3c13adb2ccfa6926a	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_ellswift.h	9148	620db468b1c13968946e9f404ceb59a3f11c895603bf1dcd3399567d7bf10e99	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_extrakeys.h	10870	7c3468c9fd192ad152c5cf042cb44938961d752fcf19966123ce6a5e3fd31637	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_generator.h	10816	1c08ef2a929eb52bff724c2d3d6e8d5a355382d76f9d510c2c4b769e5902ea6d	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_musig.h	32175	e6e2595005881a64bce69a05cba3e2715e56a7a06398c029f736cc52bcfedc38	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_preallocated.h	5966	b7283d4cc9c13bb2572dc92e69cea76f3270403baf5748ea4d91a1064b951b85	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_rangeproof.h	12175	0b816f0b669a240ec5f0e90d724f43c21b03cd000926cb6dbf9ef8791e8e8ddc	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_recovery.h	5225	3b2efb3ccf53e9f1bce93a4afae31946585958d5cba3b38fdfb1eea4c6ce2622	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig.h	8177	8185d6c6ea3234fff05f38fad4b1b6f48f0574ab599dfc87297ab9bca0abfca5	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_schnorrsig_halfagg.h	4908	89669a3ad2c904072c1177e7b783996af5d72635bb7590e02284e776cc4e085f	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_surjectionproof.h	12407	f6b245fe8234309715225fb6a3fdf3a4d20a4f5a7f395c1686debae189af07a0	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/include/secp256k1_whitelist.h	6944	12fba29620f0b7624ba1fd54bd4c7bfd8d73c259c725213d28d3a32518287efb	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/assumptions.h	4667	cc875f356cc8f7f272def71f7d7261e44cc27a18a5b668cd8192c84d238e4e68	

src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench.c	10190	832edacf541809e3cbc59a4e698d5e33a9388a7dddd55662936244cd106fccce	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench.h	4777	12fafa36167bb38d28698536cdb34bdbcf8c972eed4d3cf69e71796972b880d1	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench_bppp.c	1102	7c17a6b0fada57db18cc68b1ec183f1e6346983903ee1da4f894acb09bfab6e2	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench_ecmult.c	15692	7749b0935da862249209c8903c3cb51aa8ed1ad6e4ebc0b92e3bfbb624e9ea99	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench_generator.c	2001	6396798ee1f4a018d4ebd9e962b6c289f4a1928a870c20918285500f7972bf61	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench_internal.c	19819	8e8c24bde37d16c24fb1da91617efee49000aeb8b54a750cafbbb55443d8fa69	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench_rangeproof.c	2349	6c24fa0c6c12242da3b2629742e0baab65b24967d173fb83e94e2ce1847d8a10	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/bench_whitelist.c	4089	8f77ebf85792d8270790e1c0689d24896c2da3d7ce6f35c6dffec2909ddc4082	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/checkmem.h	5355	f367089bffa53b8998047de2f718f75d627dad549a1e5dc2a9ce9ab9f581ffe8	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/eccommit.h	2198	3b04b7fdae817984c26ce995fad4f3f6d72d5c05b9d3f78d5991c531a5e99eed	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/eccommit_impl.h	3165	6c71a9cba341b7782302741603db3ae2179b699ce2e7670cb03a749b94d114ca	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/ecdsa.h	1164	f97753416b0971133ad9d8acf9af02259931d0285a0333a9753201a353070ef6	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/ecdsa_impl.h	10477	883ad9ac4dfa224a1cc97bc2a4cc09b1078545a1632917dac2524a18ba4e53ee	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/eckey.h	1449	cf511599285faecd27ace6191eca3945d85bc852b11076a1b1727759a6b03056	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/eckey_impl.h	3439	4d327c593520c3f60f7b86d407df0870c26cf58037e4156ba8f92258304fd6b6	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/ecmult.h	2768	1e87f35efb7e7e5b6ea1d6ec22bbaeab33ccd4eacc41a9092ee518ece9237ced	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/ecmult_compute_table.h	1073	5a8ce69b18d777c3ad608ef173fd6a47c75b9c37bc6801f19ca4c44eb4b9f8f6	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/ecmult_compute_table_impl.h	1916	6082339f616cd5a628300df24e66353c1374a117e24ddd474fe6151dbc620d60	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/ecmult_const.h	1354	4705e7baabcbf8fd1853f8cef45a7f6a3075b98925d91cc209afda2c956942ba	

src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/ecmult_const_impl.h	19584	07238bcf17c3292bb8198f7f37baf068bdc9a6a57f843c439e220eb08f400607
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/ecmult_gen.h	5584	b15359caff9a96a348d23305b20b6b0f24d667d2614e3e078b0749e3aaedeb75
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/ecmult_gen_compute_table.h	672	51eb54b0cb7b49b25cf751efb2799322d73c97bc9bdd2a263feff2d2d6eeff9f
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/ecmult_gen_compute_table_impl.h	4705	9ae524743bb5871bff7e41e48b61218336b1919769132aafa944283ef1bb88f2
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/ecmult_gen_impl.h	16891	a97fb72b41a12644520c09e9290346e7bd52c48492a74d6c90613e869163f88b
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/ecmult_impl.h	34325	ce70d26cb987c9bb2a2477389459437fc2831546c9c14404b58cf7ed07f32cc1
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/field.h	15119	4a044d83637686a021448c152bcef029337af9c53ce5d70983b73fa450b58ffc
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/field_10x26.h	2338	ba9a43b62e7e70e6e18af2787bd01dfc3a57fea527cf8ad0333194211ac7d7de
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/field_10x26_impl.h	52106	6f209a533fd96e73f4c92b101bd3ca46a2f7553967ba7f44650a68ccbf4fbcf6
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/field_5x52.h	2380	cbaa91c253ba84d99efcd334c7a39906c164d5edf3cde81200a44a606205cdce
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/field_5x52_impl.h	19372	414438d05bb496a6cf4e84938bdc286a0c13902fe8ea5e4a4bb862f90176af41
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/field_5x52_int128_impl.h	11353	9f3071c7da547b94e0e6eabe0dbedc42535ea7c470f1dc891d0a24aa5d19dcf7
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/field_impl.h	14052	a0ffc1e76ae4087ecb0a38bff62924710c5094ba9a209aeadc73a291b2f7b2af
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/group.h	12243	d5443ba9dc805fb378d6dcdab02bd616ab34409b7b91ceb342ff4cee12adcf7f
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/group_impl.h	35125	663da90fb18b6fc8dace045ceb28ea8fae6e1c68081c209cf0c7dc954398f30d
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/hash.h	2625	a7f39f2ed170d70841eff7d43d9c480dfda6cb2e8ee25e7e04bf7433c27193af
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/hash_impl.h	15448	70247571d95e3c8824d6ca2e90956f6941c196a42d860083040690aaf9ccf6cc
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/hsort.h	1685	220cb669835d34c252598ecff860c93edda337a9f8c2fc07b2b6cead7d48b284
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/hsort_impl.h	4798	2861489c01b927b27d99efe173799a812434748b4addd30d102187ce4f3b66a9

src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/int128.h	3622	89ac5d91d64f857038d063dd70eec4b29b4f32aa218c790fe049137c347186cb
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/int128_impl.h	376	5683c02dfc495639da259d80624711c1c449bbaee6186729e399e713a28b4719
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/int128_native.h	581	b4c52c7ff091083f5eca9879b98641c7a38356a3c5321cbcc22ae1573e7711e1
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/int128_native_impl.h	2881	ecaafefadbde3db034d4f44678096833d2d113fd21b4ff7044095b6153ec03a8
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/int128_struct.h	229	0733b3a65719a1e9a8fb3564c71a38f2f31ca3522f40087d536dba90361b66e7
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/int128_struct_impl.h	7884	0b4e2200327f5357078ba43b0e0514c58c2b1eb217b87ebc14d95efb913b5be2
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modinv32.h	1865	774887ccd462b145434d1d6b9ae9455aa3618e791544a2b61267e18931d5010d
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modinv32_impl.h	33492	befe02af7099a871677c7b05a81302dff5a4dc1d59f19b20bcb58cc28e096112
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modinv64.h	1969	b71cf177d768beac28fcb4f0ef545aafaad16ad0f275c70ee4c60b5f7549ac41
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modinv64_impl.h	37653	1254dc7ea941fc4d4ef0a85eed80db58d9e2647a8b8d19238d6afc048d63764d
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/bppp/bppp_norm_product_impl.h	20378	13b376aadaf6ad496b82de06e57ce781c8dd591670405a50b41beb06d8b71614
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/bppp/bppp_transcript_impl.h	1552	bc8b8afcf3e985bdb355e715c2f56f109e2fb5a109f393fb0359db17b8aafbe4
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/bppp/bppp_util.h	2794	fb22fd546df224866adc6c756f76170b4aa2575cdda6872ee56f32af8079db95
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/bppp/main.h	403	20d3c6914baf5513f17d02787e51a97ecaa1670202ea5ca15bf7b09a35b0aa80
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/bppp/main_impl.h	3732	5b1f7a16c50b355846634222bf300c2887b094207b9a50f7488a2ff01b70cfa3
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/ecdh/bench_impl.h	1761	3929ead352b3b6425f0fdf94a1b2268d124cc0176929ae6ea3b4fa82abac0bbf
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/ecdh/main_impl.h	2929	3cf3b168e8048cd7b03c9b4d56d072214ef859a31bc970e5a16c2cbeba8fc4eb

src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/ecdsa_adaptor/dleq_impl.h	8527	601af2921f4ae4f9964d440e0ea755ac15276b5dc90468ee640eb58e79f7cf69
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/ecdsa_adaptor/main_impl.h	15218	30b5b37a9b7ae794427399ab9522c74400ecbb2f8cafde6dbd7352db28809262
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/ecdsa_s2c/main_impl.h	8761	bbdeb3e19eff0731a80f2fbb07c5e44a2dc8d77a1944ead679bd3acfc0d51563
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/ellswift/bench_impl.h	4399	8973d64bda27eb72a4807aae665a1a2932d624370f9d40f8833f7c8f03b05564
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/ellswift/main_impl.h	27602	8deffa7b37a0602be2b87d0378419410e6cc9683ebcf3fb033d2eed56c4f8416
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/extrakeys/main_impl.h	9961	a58762afb6a494aaec984d2e65adacf1125be5e55ad2904af0f26e490cc0d3b4
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/generator/main_impl.h	17447	cde0530ab9af0b8b1d1d80e63fc8360484578fa68f2844295fb87d90e9284bf4
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/generator/pedersen.h	904	6529271ae3f7e786df540f650164e66cae35478157b5bba6fc885cde61d3e81f
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/generator/pedersen_impl.h	1733	ad22e7826eb983f0a629b20c7bb7d4caa452d09647248a9ea54be2c9f7bfdf6b
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/musig/adaptor_impl.h	3359	e22baaaa040a80c84f0cd72a8aef3b6ba4c260a926d4039bebb2618d54b78010
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/musig/keyagg.h	1324	5349443b84525a85434ff8f92a1e7542c0982b1b783a682613026203de4148dd
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/musig/keyagg_impl.h	11171	fa67be67a1cdb0ad05fc6fd5fce158731ea499a18f1b616bd17fe3cf19dead87
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/musig/main_impl.h	452	ea8d83ca27ceafc2d119c82d62975840302871978dd7a0309f8e3bcc132ed48e
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/musig/session.h	861	c256c3742a688495205698456 40f152a22a4a8c21a96040cf0c1054516983980
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/musig/session_impl.h	31534	45c89618be2ef40dbc2006f96c6ee67d4ddabb84df2588731deb20e513c85c3c

src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/musig/vectors.h	37797	c7cd9dc3804d864d17e43c5b00335a405a3bc4a30861838ac4599a213231e132	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/rangeproof/borromean.h	1167	204588b38d00b1db3765ab4b2c25e01f5afb5f5920a380cfeca381ace064377f	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/rangeproof/borromean_impl.h	7934	0afc699394931d77dff7d20f8ef6a0a18dc5146cab67921faf8c9ec2cd0e87f1	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/rangeproof/main_impl.h	5051	5403c2086d4b81f9e85a91f2290cafd3234e111fae7adbd065a2f5eb78adae82	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/rangeproof/rangeproof.h	956	a8d6285ada07204cc46dab26087b3c61b732c3b37cbcbbbcb793a8705f0ff80f	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/rangeproof/rangeproof_impl.h	26864	a691750d411ed06b4d15f1f566db992e77f7bb075f6ec52069038979dcfaa42d	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/recovery/bench_impl.h	2320	20d13192a4aabaa52ac4c2a2da772d8aafef00773051836b25faa2816226b386	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/recovery/main_impl.h	6153	93fb7455adaa364a3c1f8acef82ba409b88a49f688dfe23ab8555510fc614c96	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h	3795	9655992c7fd214231623716a738a1928e24a86b4bcdd909f331712df800a1abc	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/schnorrsig/main_impl.h	10885	24e5cbbaaf25cfa613b42c4b0d2e146ab85188a9a262fded6cbc40024a77c440	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/schnorrsig_halfagg/main_impl.h	7995	b39f484416360cfed7d061b0e7f80c0ded79c0ed68473d377a07e3f7f434b36e	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/surjection/main_impl.h	17974	f34a06bebf853edad54e7363c122f422b72eafeb3677f52f6216e2ee08422f04	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/surjection/surjection.h	1246	1a3179eb1793a56f224ac54942c07c5defad4d904382854090e5807877d76a34	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/surjection/surjection_impl.h	5058	62b41c3676ac78bfc7258a9513676fdaeec6bef7ef90b3980ac098f08ff42b27	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/whitelist/main_impl.h	6884	2f90904122c9b814a8f765c03c3c7d50bc57affb3ae720362ef2931ad2d3d306	

src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/modules/whitelist/whitelist_impl.h	4915	3db091ee31cde1a3e2df2e2418fe07ce2f2d5c26cc63df71b1da49248efd4f1a
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/precompute_ecmult.c	4017	8085ea86390eab2afeeec8b4f952036492fb81606dcd7bb038f940924b1b8400
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/precompute_ecmult_gen.c	3864	23d1dadb40296d2752d024678e405b18d5c904fbd61d74598df3c8005ba64d2c
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/precomputed_ecmult.c	2409168	1fba19c482de09f566ed9e7bf6a526e4edb59a1cf38218e3ab413000a0e0deda
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/precomputed_ecmult.h	1585	d792b8a49759deff5f1e4092558fa9ae11d24e02695e44bee63a8f0ba8e18503
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/precomputed_ecmult_gen.c	259715	c0a7af0884b7b07962be67e9da674e0d75a8d412d116153b8eae2a9e5527d401
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/precomputed_ecmult_gen.h	973	8f514bed441bccf7e34ff05ff49d3a44e9395519a8fdcf176e6aeab7b5dbd5fc
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/scalar.h	5732	fca33c29790cc58c71be8c8918a9f15a86b170e648e3770b08914dcfc2834017
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/scalar_4x64.h	801	42aed583b28bda7982c54a6f63e50c0ae5f9c27d5f66d92ec6bf6c90e765735a
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/scalar_4x64_impl.h	38397	1526b4710019a16d5ed719ad3559427366c662fa585206b1c0cf20d4e1d8f12c
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/scalar_8x32.h	725	0614cea113aea88a47ef3b2ba8167722e850d7b2a696a4f16b03ec9a320ef1d6
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/scalar_8x32_impl.h	32567	e1e3b7acc395ec440778e3ce99d139ffc9fd981981fe60976f3c042d7388c91b
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/scalar_impl.h	12113	6ba654bd4edc0349d1d7f1dfc25ba2d73dd9f2ecdca8078e384bf7b7837d663f
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/scalar_low.h	1135	3cab20db10224ff85ee7c6d975d2968c1370ed8e96ea26b94650f6e2bac31ac4
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/scalar_low_impl.h	6503	0f47343a7d632e1aa983348c25be2fa011775208d66c6c6ddc1dba2d58172de0
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/scratch.h	2224	4af5e1ff665c0418bcc5b1c2de954e7b377f9bdfdd4309e0296938918fac11ef
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/scratch_impl.h	3677	294a55d429fa4cc9dec46ade345bbb2bfdff4055eb1cbf1ba6168a23e5e74db6

src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/secp256k1.c	34125	edf17de4216e2304af018511b54e3691021896811eb64693ae950254abbe7693	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/selftest.h	1494	d695cd5a7369b372285ba0371edb6bebdb3dd79b719af336c714a3013adba6d3	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/testrand.h	1950	bfe1b9a78209aa90d5652a83efccf633bbd874ed72dfe523f2ff6125f97ceba1	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/testrand_impl.h	5786	ac12496cdb04f5432b14dbb60ab464317669532fffa5b1de327e21075f01dbb6	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/testutil.h	4945	5a780fc16fcd8549ea4774d15beb2d36c6ad9a2ca7180dd95f84199a2863d1ff	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/util.h	16909	950f48c6bce72ccc83cce14cefb441ddd7bc1377f8a675f453d2adcc0fa1108f	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/util_local_visibility.h	402	d47d4a58f40157ab2a298151f366c5ebec8f7a1e7aaba88e36b530b1c01584ca	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/tools/symbol-check.py	2428	1aa22d8ae539b6af24304679bc92bfc0bdde47818d9b81ee92af7800603bd34e	
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/tools/wycheproof_utils.py	427	9f2c2368953eeb3c9b7c3b31a807cde76b3106df93fbb975993638d5eb1a1e8a	
src/rust/bitbox-secp256k1/src/default_callbacks.c	297	da45e3b9ba84a8289455a354a0e12d934ac515a09c0c0080afd1561c08ba0074	
src/rust/bitbox-secp256k1/src/lib.rs	15733	42fc499883f91dea548d9ff0b11ad9770a60a560c74271563e66e7e573d77e28	
src/rust/bitbox-securechip/Cargo.toml	862	6f3d69d24c02306d7c1b1ebb8d0e6f53e386217cc4132f786bdf4c14b259c373	
src/rust/bitbox-securechip/src/atecc.rs	3668	a8cbd36ebbda21ff5f525a3842e75e94fc76bf377aba97ee8c924a87e0c7f7e5	
src/rust/bitbox-securechip/src/lib.rs	3301	9d71da45b4b72c61b0f0648a6e2756ade7604486ecb0f2d331ca1896c6e3764e	
src/rust/bitbox-securechip/src/optiga/der.rs	1945	51c9a5b5999f696830104a30fc38669b68985cc1fa15ebe1a2c8717fcc4bd124	
src/rust/bitbox-securechip/src/optiga/ops.rs	20626	28f6ef019296d7087c62a4ade4089952ed70e8b5f4f1151b1f2061c495442071	
src/rust/bitbox-securechip/src/optiga/ops_fake.rs	15685	de3f330803a923572ae8e073ae4af8a057a0391c32a6499d62bd4c962eef60b2	
src/rust/bitbox-securechip/src/optiga.rs	37182	dd80d44ba7dd070e3c0c1c2ef42c318cfabf7e96eaaa9817aae6e4ff1beea024	
src/rust/bitbox-securechip-sys/Cargo.toml	259	c8ecf19222651c1788eb0599179b20ee850fff55cc34e3ab9c36952289951d71	
src/rust/bitbox-securechip-sys/build.rs	7537	3c435995bb9a2ac536ea75f6bc5d2857a14214b43e24d496e5413b4cce091855	

src/rust/bitbox-securechip-sys/src/lib.rs	311	1531ac58123bde8556099e327af91ec8e5db645bd971854fa6876e7a8d100ffa	
src/rust/bitbox-securechip-sys/wrapper.h	123	c2f4dac6ec8ac622887b6938beb8ea123b16d32fc64c151256fc5bef9159917b	
src/rust/bitbox-u2fhid/Cargo.toml	212	3e941a6766699c0e11a57e5a81779d4563e0ef4cf9b71bdc56a4be0bfbaea031	
src/rust/bitbox-u2fhid/src/lib.rs	23418	7b70b84d5579fba6237c5a176e5479c5b94502f655d63e81e047400350dc8b07	
src/rust/bitbox-usb-report-queue/Cargo.toml	291	053b288e4a5823e95b6362dfd133f63f2bd87634db7343b95c931ac75007b710	
src/rust/bitbox-usb-report-queue/src/lib.rs	10931	5a161e6b53648bda9c93dc66bbe58e5f1de2019925404073a2ebecaf474f807f	
src/rust/bitbox02/Cargo.toml	1669	b70d719688aa910b578befca33864e88ee5c4011f2c03e68cb22e3ff612bf4df	
src/rust/bitbox02/src/da14531_handler.rs	584	a4ef27ce8aa7d67aed9ec004d45235b0ebbb9cf9f6cc3f27b4f2963edf179343	
src/rust/bitbox02/src/da14531_protocol.rs	880	63a96c8132ae1e3c14081bd8cef8b1f4617c4ea1ef586091cd8dd70c0205d72b	
src/rust/bitbox02/src/event.rs	248	bf922d48b6110e575cf44bc25bbdafe56949fd54ef922730cf2b028d092735db	
src/rust/bitbox02/src/hal/eeprom.rs	942	8154b22c49dfc6aba4a2c3b9940a0691738d899e4fe1f086c35a992e9a6f6a14	
src/rust/bitbox02/src/hal/memory.rs	13523	473863fe6af3fb418ab72f97c99e8312d305a39517412a675e08e8175b793259	
src/rust/bitbox02/src/hal/random.rs	783	2e1db38e7270a448d0ec60d8edfaec3156bdca102d6c3daa149748289fa5c22d	
src/rust/bitbox02/src/hal/sd.rs	1306	7ab3174a13fd91733d38d08b24bd2ccafbcf1243dfecaa37d93527ad095b77aa	
src/rust/bitbox02/src/hal/securechip.rs	11172	1cd7cce092509b4025fea0317afe9aa7ef2e59bcf9958b3c20d0de3d51bf0a94	
src/rust/bitbox02/src/hal/system.rs	1477	6b1b1d5546d581e3ffc1fe9d7da92e52d312e0078cdee9db506dcd2ed04ce170	
src/rust/bitbox02/src/hal/timer.rs	2636	85c0bed052cc43976f4c401ffdf5d83bfe1d7d15b2b84a6c2aa0d80984b3b5dc	
src/rust/bitbox02/src/hal/ui.rs	14681	8e7d3ad2359f392ae852b68da7d6723f831adabedaf5883e9d00fa8012d81a96	
src/rust/bitbox02/src/hal.rs	2058	d6915485b75c894ba22b5a62c8b90d521f53157ce6c2d217d4bead82eccce5a8	
src/rust/bitbox02/src/hid_hww.rs	321	724870b9fac503e4e30a9dcfa7f3bc066419776ff26fdc2ca94f88fb72aa312c	
src/rust/bitbox02/src/hid_u2f.rs	321	8643cda0f6b2d05d969ce9173354761b2264745d3c4b65277a5e8650389300e1	
src/rust/bitbox02/src/hww.rs	101	d4cbb84bb30eae55a5d2a57c3b4cd3d1581c465305f0a7aa8228c62e6cb09d75	
src/rust/bitbox02/src/lib.rs	3431	b6e93585807f0e85bf91e713932d289f18759f3e9f90871f3e1e2d24030a5c0a	
src/rust/bitbox02/src/memory.rs	29086	d3ad0bad7493c2002c330e45c3fde403fb95d37db43f9cd3f9c7ea68fd9c6548	
src/rust/bitbox02/src/platform.rs	387	d2ecae56b160e9760f16ced2e9a17174b270fe43e0d7b714358bfa17f4f50e0f	
src/rust/bitbox02/src/random.rs	1425	edcc95719949a75adde6a34205951544cd8960c8f8742cf82d7e1a24480aa218	
src/rust/bitbox02/src/screen.rs	1206	6b5d0d4eab685a82c934dea778c963e411c91aabd8a62f943931d734c566eabc	
src/rust/bitbox02/src/screen_saver.rs	686	f08978cd4f9adc7285150e7fc806a61df24fda669027b9137356b44c113e3948	
src/rust/bitbox02/src/sd.rs	3633	332eed271f3a1b655dd3bca4a6e4de60a011e46d9da11fdabfc70e294110065e	

src/rust/bitbox02/src/securechip/imp.rs	6338	83020355fa265b9e4e37c87c4f36c5909b62d628162b1982f065c6ed6b54d55c
src/rust/bitbox02/src/securechip/imp_fake.rs	3211	330cd9da55af9c84a08d8f350e84e2fdc8c1c73ee96ea77c91681efd17c22af7
src/rust/bitbox02/src/securechip.rs	259	74f1549267b89c45abeb33faa7b537897aad0c8fc1257ce160dc50e960a24cc5
src/rust/bitbox02/src/spi_mem.rs	1334	f62ac7882ba064c5e08ef0f6e6d6fd466bfa68adda78f14a34c55f5772cd8049
src/rust/bitbox02/src/u2f.rs	198	1a80974c4b199c6418d90887e5895b2e89ea249936bfd652315fe8dec147305c
src/rust/bitbox02/src/u2f_packet.rs	463	1de6b32c0488616e6343e5ceeb4131d1621a35824139984fb1510d09c052def3
src/rust/bitbox02/src/uart.rs	822	12c23b2e2d988d49d67a3efb9647408309d40fa0e95b1f4054090e6e321d0d15
src/rust/bitbox02/src/ui/types.rs	3770	09f8ea3ffa8ca24ade9793ffd205a1eda3ac3216bb291daf72fc3a770d29572c
src/rust/bitbox02/src/ui/ugui.rs	72	c3c17298da8d70199030dfc0c054cd953f08460fb924f74ad9dab7a9dc2860d1
src/rust/bitbox02/src/ui/ui.rs	32463	df7dbba10ee96a9ded04a6cf9891b49bb239d8374911e637073d315b1fdbe02b
src/rust/bitbox02/src/ui/ui_stub.rs	2885	44a7c3b57ab698f176f1a6dd661d2b4fdd8e44481e70f3b52da95b50e2b8b820
src/rust/bitbox02/src/ui.rs	632	29e754c46c1185d2fefa2261446a4d6cdf656ff2f2675233ea373e46fac5d8fa
src/rust/bitbox02/src/usb.rs	153	965aefb4815cdccdf10f5a2898efd9506e6fff5de3ca33e5ce9234acfd353544
src/rust/bitbox02/src/usb_packet.rs	912	b54e3d1f804137366c10d343e0b69b0433384ec1ff156e15d108ff61c0362eba
src/rust/bitbox02/src/usb_processing.rs	1240	60de0f4120276030ec4689de0b3e2df8e83779f34627eeee87533081970fbaa5
src/rust/bitbox02-rust/Cargo.toml	4607	70708f2472b1cc14d6973c14214edd75afdc82ef08293b74fa3212b721ec253a
src/rust/bitbox02-rust/build.rs	2374	0fdd53625ca0ef342a07b6511534ccc0413dfb7ae06c455dbc58e14661a39798
src/rust/bitbox02-rust/src/async_usb.rs	16034	f24da5058d7c408f62069012f5f2463b1c76a5421a3d4e370c556121859da605
src/rust/bitbox02-rust/src/attestation.rs	2902	c20b01cb077b8baaa32a6ee878c0fd3038266ad5659829bdfbb1ec9bfcb19404
src/rust/bitbox02-rust/src/backup.rs	21345	4efaf420eec850fec73301ebd5a101c28acd12a1fbcad82d7a2cd32321a34fa4
src/rust/bitbox02-rust/src/bip32.rs	13090	69904362ecbb2911101474f83924b9dd5cbd17e7ea748a733573e9d09d02d7ea
src/rust/bitbox02-rust/src/bip39.rs	10472	3066de8005955ca3128da1bab484d55ba24d368a5ea113d36f1d89123ffe47a5
src/rust/bitbox02-rust/src/communication_mode.rs	1970	f0301d116abedb634542e247de509b5893e48290c851a420a948cf3db219fff8
src/rust/bitbox02-rust/src/general.rs	367	51fc5e6dda75ecf8de8e25c673afc1bb44214079f9fc848b5f1b733e7191dc09
src/rust/bitbox02-rust/src/hal.rs	120	18d3943a548c4b5d51e366ea87f01d61e419958ea8f91002d931ea04c1da10cd
src/rust/bitbox02-rust/src/hash.rs	1517	4b667a9836bade2671d5c320ba17c8644e245b613d4e6cee8d5f155697952ade
src/rust/bitbox02-rust/src/hww/api/backup.rs	14916	5374eff995386d703bcf4520b299484ee5673049bcb93f968f49c5374c0f0657
src/rust/bitbox02-rust/src/hww/api/bip85.rs	4647	ca089a3a18442b292edfe900b31b8c59f600cdaa5e9cfb287c4683474a39834d

src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs	41098	e15d530066a769341285a06cd933c9a5bb5fbffee95c9d6abc10ed581713428d
src/rust/bitbox02-rust/src/hww/api/bitcoin/bip143.rs	3095	1f374135c58bae745e88d9500fc29164ac723d522752bf67fbce069a45730684
src/rust/bitbox02-rust/src/hww/api/bitcoin/bip341.rs	5494	7217fe9028053addc3c36337b2a43958d7e56a3a4d8a92c1e630dba3301fadfc
src/rust/bitbox02-rust/src/hww/api/bitcoin/common.rs	26882	b8a6f3c01008c0631fb04cbc9b8f389c6cf30de321502f994110143f0a7eab56
src/rust/bitbox02-rust/src/hww/api/bitcoin/keypath.rs	18724	d5d08ddb0cf20c3ca0497a723a08d748f863118b90d170ddf1a0d308a7d39801
src/rust/bitbox02-rust/src/hww/api/bitcoin/multisig.rs	42734	cff6a02dc53347895b0f6c7562c316eb354eba5a81ecaf8cf111da6f00e8a88f
src/rust/bitbox02-rust/src/hww/api/bitcoin/params.rs	2291	fae19e0b1a338960fb97055efcd6bd6c1d1572962faf960644cdf239592341ff
src/rust/bitbox02-rust/src/hww/api/bitcoin/policies.rs	82001	abc68ccee396717d58844920139f2d29bb75a4fabc74f976897470f0232d4b91
src/rust/bitbox02-rust/src/hww/api/bitcoin/registration.rs	12363	338b00774cb29f0dbfe5a480dc6c944f70fa72d112e976fd4e9bb873ad1deda3
src/rust/bitbox02-rust/src/hww/api/bitcoin/script_configs.rs	5506	17594a528cf37f0398f0deb18b16e996060fb670dae8520de29c243379c7c170
src/rust/bitbox02-rust/src/hww/api/bitcoin/signmsg.rs	25339	7f2d957f821c1930f16dc8c1de5982ece4f50b3468307cdbc8925ff34b21a979
src/rust/bitbox02-rust/src/hww/api/bitcoin/signtx.rs	138278	195b930b906cd84cefc9bb4ceecae73268d702b22b019f80b18992d56c6a45d3
src/rust/bitbox02-rust/src/hww/api/bitcoin/xpubs.rs	6970	4bc855177239b7b5cf15dd545a461bc421f39555d8f7fb3a65b672af96848836
src/rust/bitbox02-rust/src/hww/api/bitcoin.rs	53465	181410dc293fae8eb397eeb72e4b0ac6094efb5740c9d8de249b12b8253e69cc
src/rust/bitbox02-rust/src/hww/api/bluetooth.rs	14999	54fc1a9ca2d9412a3708551436f0d0b2b4c93ff2aaae3f51deee8c3a77c4e539
src/rust/bitbox02-rust/src/hww/api/cardano/address.rs	25289	2edeeab72109e5d9c5244ae16cf22b7397aa658364c5f27466538b16deb366ab
src/rust/bitbox02-rust/src/hww/api/cardano/keypath.rs	9489	669a3af625f388c46d5718aeb4cd88476c16a2629e3d12134b811f0068882350
src/rust/bitbox02-rust/src/hww/api/cardano/params.rs	1354	d7afb92488f4c471889b2d5db2526630a474377dcbd903d85659d12d671ace3e
src/rust/bitbox02-rust/src/hww/api/cardano/sign_transaction/cbor.rs	11409	89e1d37d2bfb99b67ab18817241be589831874e5f56a5f0075f3a6bbf0bf70ec
src/rust/bitbox02-rust/src/hww/api/cardano/sign_transaction/certificates.rs	5772	039d3fb80dec40bf724d7daaf3dd4ab7c9d3f1d35b159a04503ead5a3564c8d4

src/rust/bitbox02-rust/src/hww/api/cardano/sign_transaction.rs	74982	5e94e09ed5e9e28ba9935be3120f5926f5ebcf6c4a275e79c2a25c2cc0564ad9
src/rust/bitbox02-rust/src/hww/api/cardano/xpubs.rs	5566	ca845c4de879d93b868925ace61c0471acf3041011d8bb63f916239064c2055a
src/rust/bitbox02-rust/src/hww/api/cardano.rs	782	fb2b77c02fecf5d5b0927aff56ec1ee7c4d90fdbaf3e112043b66f58508891e6
src/rust/bitbox02-rust/src/hww/api/change_password.rs	8093	df7e6672fc66ba2ce24c092ff0b58224b7181bf87fd978f501eea082a2aba4ab
src/rust/bitbox02-rust/src/hww/api/device_info.rs	3187	cf7700970753c5467429c85b88e4f21dd87f88aed83c0e879cfc899700c270f8
src/rust/bitbox02-rust/src/hww/api/electrum.rs	3010	9cf68840f7aed0f19a71e0ed7771e423a77cf662ae54ffc196e88999215249a7
src/rust/bitbox02-rust/src/hww/api/error.rs	4335	b6a13ec95227e726cf487267d3a0ed53bc9cab5b1d3860d497dee0ec97cd59c1
src/rust/bitbox02-rust/src/hww/api/ethereum/address.rs	6735	57664aa3edfb4b15fc9341ccf65cc9c7da221fd0a59b988b4a4b2bbd05d42d37
src/rust/bitbox02-rust/src/hww/api/ethereum/amount.rs	4447	78effd3b340e7034d0c5bb3332151d9fca0b944c312f2235be007a3bb7d0fb74
src/rust/bitbox02-rust/src/hww/api/ethereum/keypath.rs	5510	e7129e999976e13aaed04d41604ae3e7d65b5d3e8c6da7d5ece9063554523369
src/rust/bitbox02-rust/src/hww/api/ethereum/params.rs	6318	6d45acd07d99ea9e40064d668797b13f5b24947beca278a8ec67ec96a63c68ad
src/rust/bitbox02-rust/src/hww/api/ethereum/pubrequest.rs	11492	1fb38f29c8acc366e5e28ee31505eda46b33d29abca6006453cfb3aeb5fb3614
src/rust/bitbox02-rust/src/hww/api/ethereum/sighash.rs	27142	cccb0d7501d1db215c00bbc9586b3d873625b63606d2a072319eb5b1f7408e8e
src/rust/bitbox02-rust/src/hww/api/ethereum/sign.rs	96116	1891a4b2d603dcb45aedd9b2fc409ddaf9d27a813264145b7b9b3cbda7001b46
src/rust/bitbox02-rust/src/hww/api/ethereum/sign_typed_msg.rs	117444	890c26321761c405f6ec0b238a67ecfa3aa3ad9d2c594f2bcbb85487e57020de
src/rust/bitbox02-rust/src/hww/api/ethereum/signmsg.rs	10476	b4de40b5a63033ec24e52f0077c80a3a73a9471fdc6d10d5289322356cd626cc
src/rust/bitbox02-rust/src/hww/api/ethereum.rs	5189	873ca1378407cae796dfac9032da1bc6e980612f5f2846592ff2d493384071bb
src/rust/bitbox02-rust/src/hww/api/payment_request.rs	67439	c76bd2db8e0205ee3cf3e26cb00132da08975241342a6f88939966e2c4a24684
src/rust/bitbox02-rust/src/hww/api/reset.rs	2174	44fa3b048c424a65d24d853cd2fe6aaa805c37b6f23b2c7bc0bca915c98145d0
src/rust/bitbox02-rust/src/hww/api/restore.rs	9331	3afe08e63f02f6e0e547bced73824bb37800e52551da477c4cc95f1f2925f307

src/rust/bitbox02-rust/src/hww/api/rootfingerprint.rs	1540	bec5a60dbc30d5c9fbc4b387d121c35cc82420262422c85693f79b65d6c07626
src/rust/bitbox02-rust/src/hww/api/sdcard.rs	2540	9ac1af0cd1531ed86ab3dee9b9ca1711573ef131617453faec1a161a3245afc1
src/rust/bitbox02-rust/src/hww/api/set_device_name.rs	4581	b336413a7ca7043f89df8cfeb781e7c602739c44b6d17fd21811dde8b6e3f0f4
src/rust/bitbox02-rust/src/hww/api/set_mnemonic_passphrase_enabled.rs	2612	c1361dc350dcc62f74137349a37b5190afa02dfae0952e622569b8636f8b8e60
src/rust/bitbox02-rust/src/hww/api/set_password.rs	4915	34eb42602dc38d169e1994f8e360c960562739af09700e633fa6057dc09b4a05
src/rust/bitbox02-rust/src/hww/api/show_mnemonic.rs	7482	a5180d83d601f5774114f1cba451a2783358e952e52cbbc421a1de46c81f4f13
src/rust/bitbox02-rust/src/hww/api/system.rs	1785	a0527f375722906e809a1410e5cbbd8404ec3d19bbd8d4d278e675a8b69e2531
src/rust/bitbox02-rust/src/hww/api/unlock.rs	6665	29ec66437733d134968bb4e5b366d71cf0e47aeacbcaa8e5d157ad2fc90c889b
src/rust/bitbox02-rust/src/hww/api.rs	8874	4e88939f98f0466f8f635eac082d397f58abb032590ea8f0a29afe55741f1f7c
src/rust/bitbox02-rust/src/hww/noise.rs	5095	84f8975fbe603db802539bd6b6973505ce5938787617d4f53fa3923056d50024
src/rust/bitbox02-rust/src/hww/transport.rs	16808	371927da8fdefc6f47d8a5cdb7518242446f2340efdcca17ca0b90a5ee023dab
src/rust/bitbox02-rust/src/hww.rs	28291	2927bcad9778746803fd0da2c14bf7959bea20c137c5fd572a5e3968e3614cb8
src/rust/bitbox02-rust/src/keystore/ed25519.rs	10193	85cb066d8f6e599c0f1eeafe7aa139d6c51c0bb678c4d0308173de566c61304d
src/rust/bitbox02-rust/src/keystore.rs	83987	b7918fdd4b52ef1b088e28bf88f3442f8243d706fdf1db79c60e71fb8817420b
src/rust/bitbox02-rust/src/lib.rs	869	54ccda57e672294cdd51137f4e529824830c58457644c67d8b6d2952442471b5
src/rust/bitbox02-rust/src/main_loop.rs	6571	8f8dc3d87d7a372ba046e88c3e844c1ca14b8ff5340d715af412297fe820f381
src/rust/bitbox02-rust/src/reset.rs	4440	fc08d0007d657de5b8f162034006b1eefd42822948c09ad247aacb2bd28ce528
src/rust/bitbox02-rust/src/secp256k1.rs	287	4da9a5cfc6fee6086c57c29d11367c6473c6f2bc7e7ec31c593ef67a6464883f
src/rust/bitbox02-rust/src/u2f.rs	6712	3d6bffb5411151496b3962d2f9c5ee559fc86e060f86791f8b2ca1b40439e5e9
src/rust/bitbox02-rust/src/version.rs	166	49bbc061dde4aa1a081e6e79d3e0a65ef4067cadfeb636c0b02a4aa6f311a33b
src/rust/bitbox02-rust/src/workflow/confirm.rs	1449	84c4285756710f56d7ccabd75246644e49ae186c2d1d3aae6a4579911550bc02
src/rust/bitbox02-rust/src/workflow/mnemonic.rs	34778	2257e6aceeca61f3939defe9bb98fa092f5ce6bcd4a424b84aa298af40037831
src/rust/bitbox02-rust/src/workflow/pairing.rs	1690	345b1b3d5c9873b12eff6f03f40d279bdd27dd077daeddab88199d49874315b7

src/rust/bitbox02-rust/src/workflow/password.rs	6278	f0a995ecf8584cd5c58cb37f20686d39e9654c5cfa35289d267b3ff939c3b019	
src/rust/bitbox02-rust/src/workflow/transaction.rs	3554	6451402352fb8ff628bba81fabd478aec289f32385735b8b68a1372b6fc02c05	
src/rust/bitbox02-rust/src/workflow/unlock.rs	17747	065962fe57d9cb6640869677d8f5489bc0fc188a060f746dd83b562f80805fa9	
src/rust/bitbox02-rust/src/workflow/verify_message.rs	13559	4a2c0f3cea125caeae7d0371c7629ae08b03f3cb6d76f183cd6d11385e6fe846	
src/rust/bitbox02-rust/src/workflow.rs	296	885ca432cb372563ae134d554254bdbd8b914a880c12c5e15517df1b8fb668e0	
src/rust/bitbox02-rust/src/xpubcache.rs	11790	fcc3f7f8340292a2d682a108772c490f6d41827c31f331a0edd32f4d7fddfc8d	
src/rust/bitbox02-rust-c/Cargo.toml	3932	2c730e8b213a5d8e3a381d0a5e4c2745e0ba2a12344883db570fd8d70ddd061c	
src/rust/bitbox02-rust-c/src/async_usb.rs	2063	47a9499822950a6254d095bc1561302952fe6335fa5563cc400b79426b97978a	
src/rust/bitbox02-rust-c/src/c_alloc.rs	905	471866858b22495db4c369feba9db9b5af816cb01d5e4c87f0f882573f1fa697	
src/rust/bitbox02-rust-c/src/communication_mode.rs	378	c8d296555831770fa32351f31bad19787ed3693d366c9f42f2e4c9d9515e971f	
src/rust/bitbox02-rust-c/src/firmware_c_api.rs	5436	029d99f0ae93aafe5fa6331d4ed1787be0189241340a75018052958599fa4358	
src/rust/bitbox02-rust-c/src/lib.rs	5125	c872df632ac63a1f4735660e6ac132b661f80f0939a38c07914203d6a3ac31f8	
src/rust/bitbox02-rust-c/src/secp256k1.rs	6624	54c1ff4e1f9ed19a0c92a26d1f83cc8a2fea72509e0394ce4ed01973e0f2c3c6	
src/rust/bitbox02-rust-c/src/u2f_c_api.rs	9830	f9dd4d7daa5fcf8b4361292668bfaf314c9e817e2a6d8bdeb8f22e9d476ba5e9	
src/rust/bitbox02-rust-c/src/u2f_c_api_stubs.rs	781	e7f0952fdbfff2c7a2f00c76c01ad871c4ce0bcc8d783a445321b1f27840ef5a	
src/rust/bitbox02-sys/Cargo.toml	390	4492ef51f8a188413a4ee87a0dc30f58c2cd5a28f890a631480558672e3faaeb	
src/rust/bitbox02-sys/build.rs	21680	6370b2397f94e9e9531ae43ecb9dfba9e819e1d7f0dae4c23a3e89d8658a3a07	
src/rust/bitbox02-sys/src/lib.rs	892	8ed5a6852000ff1e71639e70b0d8243624213c16d85a0f7b64aff02ab41a6e9e	
src/rust/bitbox02-sys/wrapper.h	2101	ce7ee138f934a422cd5a49f23498dab908a330bbc1ebba755c99bb1832c385b9	
src/rust/bitbox03/Cargo.toml	400	6927165100e35a7e5b2e199208a4c7db809060ea479f375649961c1658d8c3c3	
src/rust/bitbox03/src/eeprom.rs	792	ddfb8813943a33930450b01e2bb1616811ee5ba688a51a5cec5f37ed4bc39ede	
src/rust/bitbox03/src/io/touchscreen.rs	1866	ae775069eb670b7d1f0e62545590e920d12b10a817b7e031a9829b2cf360b471	
src/rust/bitbox03/src/io.rs	61	2953777db481a5a05c64b72d4f3a9ba953aa2f72a9b38c2ffc270b50c2a644f5	
src/rust/bitbox03/src/lib.rs	2578	38cd67b71d70f90fee279114ad5d1398c695eaa859950b85d5a022ae358368c5	
src/rust/bitbox03/src/memory.rs	3894	4245c2ea7d8b4506f821543c2b4f92c9c03b7da8c6575f7d6ac344f07b592284	
src/rust/bitbox03/src/random.rs	259	19138960ca09317cf42a71bd818c0171ddd703c0db9c1ff6a0019157bd492420	

src/rust/bitbox03/src/sd.rs	727	6d7d16479a0cdd1f9fbd270ded12f5d8231208e48299c62bf1086112e9da162c
src/rust/bitbox03/src/securechip.rs	1946	42c32de5b793c2c21ef8cd188192fb821c778cb6b47e5c9d37eccdca3452862c
src/rust/bitbox03/src/system.rs	446	d748f74fd561901eeec5b1e43bfc50a30d3515b57090f468f786009b394d3613
src/rust/bitbox03/src/timer.rs	239	71b861386ab450012cb83ff87a1a5799dd4ebfb234b3f7781a65d5872c815409
src/rust/bitbox03/src/ui/choice.rs	3868	161d493f0600ce9c57d8ecdeafde6066ebb5ecd02e94c0e201c8d69be5852779
src/rust/bitbox03/src/ui/confirm.rs	6097	38396a0ad70cbc306c39df4293d497916bcffde5674b5918c7f7b81f385625d7
src/rust/bitbox03/src/ui/demo.rs	3421	1d9858404809ee7f4b0d9bcbba9b76fa7360386db1064c7784a6b9bf45dbb3b6
src/rust/bitbox03/src/ui/enter_string.rs	98270	3742ca93ba5aeed6c2a0a5d1a1a4d0f0797c8b0ee01e5217072e1c2bd0c4d5a3
src/rust/bitbox03/src/ui/keyboard.rs	33770	8e3bf28083cf243c0c9f45f7c30675ba0655c488ef01480bf845293f26c4d01c
src/rust/bitbox03/src/ui/keypad.rs	5537	67c0c3ed07ff6dfbd20609f8f95854356e7bf107bcd00217a3fbfaf30fd5891b
src/rust/bitbox03/src/ui/menu.rs	4895	cf75ab296292809f1b3c571e2e47024268d50aa889cf81dac52a27f2846460da
src/rust/bitbox03/src/ui/nav_button.rs	8533	695ae86711d9a7bb9b5e4dbbcb96f988de3200878dae044a86b7cb9c72a15d9a
src/rust/bitbox03/src/ui/recovery_words.rs	21855	8693f42ae62696f45f923dfadb51c55349385f2ff23484a158fb8741f8ca662a
src/rust/bitbox03/src/ui/slide_to_confirm.rs	33146	f2cc8a66ddfef7be8e0d521e23104cdfb7eeba3ff1129c99adb135883a97bc57
src/rust/bitbox03/src/ui/status.rs	2746	28cfc602ea285c799d834462b8013ee85bd14e11a0eb94f6473211d26ddee11a
src/rust/bitbox03/src/ui.rs	14682	1d353457089c94e7747b248670e64a5203f4407e0aee997d325d5afacf872e34
src/rust/erc20_params/Cargo.toml	309	9e7f84d4c08350c9fd01151b5e00bc58ae558fb6b612678ffc0b5ad9a81ec76d
src/rust/erc20_params/build.rs	5606	036ca79ff51cde8f3d84746440840185f617b63d2068367058c566003d0651f7
src/rust/erc20_params/src/lib.rs	7510	2692c60e866fd7ef4fbc5c3691b101e43f50255e99c3fe8d877fdf3211c24b5e
src/rust/fatfs-sys/Cargo.toml	253	8e99c5b88d0e76f8c315afb2a1b7cd57da52bcc40e673b717b7d0a13e9dff266
src/rust/fatfs-sys/build.rs	590	19030cba4386dfedb0e812698684f5b7f225079e2c9e3218fe44886a59b26e63
src/rust/fatfs-sys/depend/fatfs/documents/res/app1.c	1135	17f99c12e7017ceaf25734165745b5b43b734fdd6fdabf5c8c6a01ec2e778aef
src/rust/fatfs-sys/depend/fatfs/documents/res/app2.c	2236	11c9bf944aca5e55c7642225e1a59f5998a3dc43a302b6f1c5310e9a4ad1433a
src/rust/fatfs-sys/depend/fatfs/documents/res/app3.c	4133	e5b1c71593750cda3cb11ca638f75d55e23578515fcebcc7607cb66de4e21bec
src/rust/fatfs-sys/depend/fatfs/documents/res/app4.c	10567	11ff9994c90bd50a04754f6aae344cd2863b2de142d401ec86f81d9d99232dc9
src/rust/fatfs-sys/depend/fatfs/documents/res/app5.c	1411	ed551e157dc26a5902e59beba3dabe3052d1cacdb806ccfcefa76a519c9893cc
src/rust/fatfs-sys/depend/fatfs/documents/res/app6.c	1974	209a05020537c1a63cff1d5a4c48a8510ac05fddce400d2e7f440e721c330b68

src/rust/fatfs-sys/depend/fatfs/source/diskio.c	5385	435567bf00b4607a803339ec31d9402ea12536cbf65bb65ef9301027f7d38836
src/rust/fatfs-sys/depend/fatfs/source/diskio.h	2733	1c2b29f53e061c55d43b2e412c3a61320b2621bac7561a05e275514baa44785a
src/rust/fatfs-sys/depend/fatfs/source/ff.c	266970	58ec7816a45870331b2a9c0eb977db19d1f0d95d7416a1482df07675a402c8a6
src/rust/fatfs-sys/depend/fatfs/source/ff.h	17365	42920ae4d6080bfc2a72d92a9969b5e94eeaff070e5e71d90da9ab6d7d8e158e
src/rust/fatfs-sys/depend/fatfs/source/ffconf.h	12714	6dc2ff2adf09b990a5b91eb253543061bd77ed498f92f4887837b35e8ffa5236
src/rust/fatfs-sys/depend/fatfs/source/ffsystem.c	6172	4ef367d61c6060d2bedeac57985ec293cf29d33faa98d15fbd78c5d63260eac7
src/rust/fatfs-sys/depend/fatfs/source/ffunicode.c	1974995	3f1632c1f1e64b0b683b0d7e51341b9870728f398604869dd8211f1819509a8e
src/rust/fatfs-sys/src/lib.rs	11	bdf6c7cf1fc01f69cc1d4e52948d1f246fdd310e6323d5dd1efd23f1daf382ef
src/rust/rust-toolchain.toml	178	adc23c7a875dbd0c12bae71d2e35f74ef87baf17d0660dd4ec245ea41d14e5ff
src/rust/streaming-silent-payments/Cargo.toml	404	b7a6dba7cfde137b16a86d3d65157dc965d70835aa6f6ee6cea66dfe2d23b044
src/rust/streaming-silent-payments/src/hash.rs	1776	ec49885fa0902fafb0b1bbd6ac6e4444aee0a3edb622adbd887f23792b637219
src/rust/streaming-silent-payments/src/lib.rs	14461	4e33062aad0ad48284857ac80a9b626078474c94ab7901ba3f2c17eedf925488
src/rust/util/Cargo.toml	878	543d8a138679badb32951a85c9c276ce05fd7ba1ca05952148f1f844e69be7ef
src/rust/util/src/ascii.rs	2089	fde025eaad4c2af6ae324939a7fdf659c09af403649596354181de113363b640
src/rust/util/src/bb02_async.rs	1686	6ed61692ba8092d58add3b157ed5eb9cb1fa81dd448f53cf5b56ba7ae025cb21
src/rust/util/src/bip32.rs	1412	457688519fc8b6a097ccabd27c189b1f6ed218437d8b478e029459d58c781125
src/rust/util/src/bytes.rs	7737	9e7f5104142dc0fd492bcaf693ebbb2c4bc98a91f19cf3c929b4185f3ce4c5fa
src/rust/util/src/cell.rs	1286	4fb9bdf81aa82bac48d10dd8ba8b2443e2fd5d523ef6b6275cc09ac5d614b975
src/rust/util/src/datetime.rs	3468	741a7d3fdce76a0a30d332b29f73f237691783190c06f9c3922b0a532fe0e939
src/rust/util/src/decimal.rs	4784	01a1b5c401e594edfbbe05a5261deaf57841f8a63c5f762238f07b580db22509
src/rust/util/src/futures/completion.rs	3750	339f12b2cffad430257aee544aba322fdcdd803a23b52dc443480e22f0988980
src/rust/util/src/futures.rs	60	8fdafb7122f20d1b5ce988e860031c57512db4fcc1d0a86d01e53777f5c4b0c1
src/rust/util/src/lib.rs	2166	dbba9c609740021556b8cde831364aa22ae8415addf76f481941ee34c09c2de9
src/rust/util/src/log.rs	4249	d9b518ed3e5292c284bf90958e551320c39b18c1bf88f79094dedc7ee3698be8
src/rust/util/src/name.rs	1558	998da04d404cfd53209df659581886544d34ae1a4c314a0c044086f7b837e883
src/rust/util/src/p256.rs	3298	7662a6309089b842bfba84fb5a187419ef8588678035d9f9766bc456d9ab0185
src/rust/util/src/sha2.rs	11704	ea2cb924dd5d2189531c80450997fba14624ba4244fc2063ded4b5bf7f4c5648

src/rust/util/src/strings.rs	9623	bb60b4f1763c9444ff1675b49c130c87ed3394b6ef479589abb168e6ac86b811
src/screen.c	3476	cfe33ab69a62ae78bf8bd1a52d007ecbe0a2d8799f1917ce5052f259acabaeb5
src/screen.h	943	35e89291cfb88bbc08bcd4ab8b913e406808ac84fc5a112743ad60eb32775262
src/sd.c	10969	8dcf171b7d515317a15e17a90a6fb1073bb4175e9eb3561e415aad93c5de6f31
src/sd.h	1898	c91c4fd60a64f13e33871264bc51668534a51576535f4b39378f24b2f1be79eb
src/sd_mmc/sd_mmc_ext.c	698	79fb100d033d6068fc812405f98f56ebfdc9536fc350fbc80bec1629e1839fe7
src/sd_mmc/sd_mmc_ext.h	206	6b4c4b41838ce53d4723c1566344eb30dfa8292e3ff033e33dbc6bed601345d2
src/sd_mmc/sd_mmc_start.c	430	ffd9c7aa3251780e2374147433d3bad4e01b670ad9b0e6eb0c0c5f59a4fa5659
src/sd_mmc/sd_mmc_start.h	51	ec31f931047363fac9aee9778d3afc0b523683a9d5b21701c424f062a30a3848
src/securechip/securechip.h	2253	553fd604572b88af6d037102db994a94ffbf5bab836db0783d439b3d70e6f9be
src/system.c	1093	b6ff1e7ab0ec14f265f906eaae439977df28d9b9bfe7835aee62a71dfdd792e9
src/system.h	262	b2188c4cdc59fd46318d22edd41cf2ddfbe25023c3226490c7b2716ca64db2c0
src/touch/gestures.c	7780	15d289b74f8d1afd120c857425dc1b5614c3aac3bd767564056cad032ee00228
src/touch/gestures.h	1036	f1431c6bc572c1ff0bb639b596d83bfbda72e30440550db2f7af5b943e45650f
src/u2f/u2f_app.c	2770	0cdd0a8722f62744d8f89ef5d303b007836f4fb5ab5c70086639149abf965394
src/u2f/u2f_app.h	1429	4922af347b06b58cf350f854f0e3d824d34c4d6ebbd5252ec37e7dad3bb3877f
src/u2f/u2f_packet.c	5265	b255472c0baaf4fa28be35ceaf7f367bfac3c144800080815b0ace821daf9c55
src/u2f/u2f_packet.h	1048	62fec15e5a9779ba29e3876e6979f65a3e9918c7758a467c3099028b64fb8f0e
src/u2f.c	32289	5f6bfa1e7cc4b94edf3c23fcfe699c74c484856841debe934de3f86abef08ce6
src/u2f.h	1076	85ee65d73887ef958af72a5495e5134c5619b2db53a1817379c7e90ae2af84e7
src/uart.c	3229	8a2f1cc8191b42d45b03d0712e73a238b6c1de7bd126d8325cc1b7df7fa123b3
src/uart.h	545	e43783cf909e54c395b4d4f7b9f8059ec4ee8c99a080ed4acdc9fff84b2c2296
src/ui/component.h	1821	a07473e0ffc5c839fba5d583fa2a5d543facc81064f087f04b3d79fd2605ccf5
src/ui/components/button.c	5022	cc451912b4ccd09fed913d03fd0369ad53acaffaf8778e24b9c02889046fbe47
src/ui/components/button.h	1399	3a1736c1bd6fc90a784a73bf31d7afa16714dbc5c16996049f9aaf1b65d88408
src/ui/components/confirm.c	4677	ff7fc3bff893ce08e68a27c1a73c3d21a061d03b574d7557dff7ee86e52cd525
src/ui/components/confirm.h	1636	e8bed0d30352f14a1301317974040fd9a631ffd1193fed5e1e2559a86a91b060
src/ui/components/confirm_gesture.c	4737	b74d570fd5e75f68404265f705fe36fdfd0b611122732ec21c458df31dd8edb9
src/ui/components/confirm_gesture.h	493	447f87d9695c6941ad76066593024fd4128ccf286db7ea105b9be27031c09c4c
src/ui/components/confirm_swap.c	3118	e54acf522bfd2eea84a76be00993c645a92995312eb3b385fa97f68215f76fa8
src/ui/components/confirm_swap.h	686	7a4a9ffc3cd075256c6135bf9436a36e4b0e492eced8187af5d9af66f17d01ce
src/ui/components/confirm_transaction.c	4813	810d55e494198ee188e15aa696d2354c6ef6041fa79181d385dc74d96cd37643

src/ui/components/confirm_transaction.h	1200	61f867ffb8af1557dc717e5b667f5f52ed7d4ce2255ec8f2ed591f6dc5fa0936	
src/ui/components/empty.c	679	cd1c8fc5e607e841c9d92235e2ebcfba6c8cbcd6425cb39060f2d8040ef34ffe	
src/ui/components/empty.h	225	6c17f7e1f19f22a91275f7c6093def976bd7de8c997ac927ffd7fc10a1705656	
src/ui/components/icon_button.c	5054	ecb4e2b2970c787c8db7deed590b4633ffa6f100c7b86bae5d7520e54b481bea	
src/ui/components/icon_button.h	616	4ee6d9866e86b07ca04465d3164b8773d83bedc6752cb856ed25ee1652a2eec5	
src/ui/components/image.c	2682	9000e883ffc7ddc70e34bc49bf145bc31eb3af716ec6ef9b79e42fdbbcca8b38	
src/ui/components/image.h	753	f749e5a75baf96d686d30acccfb1b1c4d739353799d340ced306c8be48dac2e8	
src/ui/components/info_centered.c	1662	be0d954ae5733ed1ea1eb5fe759236704a98e246c45f9b067719068d76b8c956	
src/ui/components/info_centered.h	550	2cc0857e51f6966a3225bb5796e8e1295f8d5bd92154a16244d548974617bb95	
src/ui/components/keyboard_switch.c	5153	4ee1d5a0b16f35b8e986e3d16cb50086340db1c5a665c4fbf79f2e7e7938c536	
src/ui/components/keyboard_switch.h	958	78786a829330eaf101021614d0f9aa94d76319965e16ef37e165393a0a71ee58	
src/ui/components/knight_rider.c	1836	e227df5a99bfb0fb90159492462438e4871b089d96f676ae0cef2194db397a0d	
src/ui/components/knight_rider.h	425	5b38b3b1f8a035e4446c84eda0a9a305cb80b80f1d7ef7bc89208213cb6af382	
src/ui/components/label.c	8871	208edee08788d00a3ccdde38fcb505a6279b0a90c7cefd00bafcbd803412e23d	
src/ui/components/label.h	3426	f135077b533da03cc075498e897570a3f0161bef23aa48864becb5977167bac0	
src/ui/components/left_arrow.c	3444	09d2d637d63f8a5c6c61ec0c6276d7f2f6fa97b23d58a6582af3f21569efc5e0	
src/ui/components/left_arrow.h	511	3196849704e2eba89ba6c5ea85f0ea13127dd1081445818ec9fa409d046730f5	
src/ui/components/lockscreen.c	3456	199c5bd8d11f4aece24ca0ec834764cf799cb2db3175fee5bb89ece21f7c5db7	
src/ui/components/lockscreen.h	223	900e7f56572d6f06caec37d075aa969818a165ccaa7c5bc4b917362560dcd34c	
src/ui/components/menu.c	9273	21ebdaac62d659f4a8019c82ab870e764626fa348923a3231e59d111ed115ef1	
src/ui/components/menu.h	1345	706c6ff3f2345513b3c10cff0806c5b85b27cb4a35c087d80716fff790912a99	
src/ui/components/orientation_arrows.c	6207	71686192af175d7e295191e46879b26be6b44c59ed696329cc1530467fa608b5	
src/ui/components/orientation_arrows.h	530	1bf76607ee6050cffed3efe2faa9a9e8ad81190411f803ee0dac28d273ab8544	
src/ui/components/progress.c	1648	947e90bc3a27026e979ce9f48f5ed71c30d9e62e11449c27ef72dc60a3eebc37	
src/ui/components/progress.h	402	8232044dd9aef38e46af994c30c99b30303d61f2d6c8ec442d3ec45f7bfcb9b9	
src/ui/components/right_arrow.c	3650	83a10b58ca690de826c330750f2e24a09eebe43a5390a304236078e97cc1dfcc	
src/ui/components/right_arrow.h	515	8ba6ea528f44d08b61d363e81d5f65d2caca802205d89c8b18df130ef736eddb	
src/ui/components/screensaver.c	3122	474414d341c7f942ca9435d3eef090a1e4d0cc6bd507eacd20fb4e9584b5d75d	
src/ui/components/screensaver.h	375	7ec6bd40c73af2d8f8fb5cf196568acfed60b2979463df6cc9c590d7bd42a62d	
src/ui/components/sdcard.c	2748	8c319b47b9d4c4041cafdf47c7094e2ff5daadc6b7d522e977ebd6b1fb937f30	
src/ui/components/sdcard.h	389	2287aa5eced8006a9e9b407923b89841de7cc505a78bb84ec5372a51c4f3b89d	

src/ui/components/status.c	1672	56d47212b55dd91f6cc887b1a1d877ebc8a8f6d3a4f778aa37ee5939fed34041
src/ui/components/status.h	507	afa759656dc103c185a053bf5ec2643fc5a7f3df5464ab170938b6cf217467c3
src/ui/components/trinary_choice.c	3257	4b7f85ae36693ca5a89323187fb542474277fed672670351e5ea096b38868688
src/ui/components/trinary_choice.h	905	86a614340e27105664d95ae72c1579823cdacad01250dcb35aa2495a2dcf477e
src/ui/components/trinary_input_char.c	14903	bc419ea37ed53cfb0c960c2aecde3c9b040ad7aa5e07081116e8f0eeda293ed2
src/ui/components/trinary_input_char.h	1663	ed246da425389a34d86d366639ce4a8ca5a1b8b93458de5b4a6f39c736ac6b35
src/ui/components/trinary_input_string.c	18762	c565e0b6e2e4f32072a611c257b76f29d20efb27aea23a38c26672a6a14bbad9
src/ui/components/trinary_input_string.h	2123	d7c341feef38c0723581490bcc465a243460a5085792d83baad8f32e17281d0d
src/ui/components/ui_images.c	14893	38788e6b993403db59eb8b9675c05c5b5f82ef12f9d059491b98dac29e3b1c36
src/ui/components/ui_images.h	1814	9d0d48dc2a867a42c1c1f8935d6964906f18c509b5183d9ef5f18b8e7027a6e2
src/ui/components/unlock_animation.c	10949	8cfedb0276426a8b6ad17cc94382d747285b74e95c820cb0b8a1f9cd62322bd1
src/ui/components/unlock_animation.h	269	031cb5d1a4a989b037b9c91785b7128f7e4a02621c4bb72973eb80e60ffd7221
src/ui/components/waiting.c	2901	876058da24e77fa58c9f4799480b7def7ffcd96bc3b80d5705fc74d140ba8fb6
src/ui/components/waiting.h	611	0c497151feb5ed87172f218a79524b725ba06c6b8ad2dac99e9123988313514d
src/ui/event.h	563	ceb3e26bfa9d5441f571eb846273ecade740f6b1353311ce55e990dd5e659779
src/ui/event_handler.c	1520	7e183e8ce31c33254a45ec1a02e2fd7d3fb91e69bb1cef7e5e190030fb08dea9
src/ui/event_handler.h	267	43ec16c77d45de807db6cd8a50463e1fc7bc4a79443493973eaccd70ce019467
src/ui/fonts/inter_bold_32.c	220230	a1b38b3f0de200baeff70ca8d4342fd1e003c319b455b3bd8797a524470ada2a
src/ui/fonts/inter_bold_48.c	477264	8f262bd976dddec9c9fc69fb6cbedad692ef7fb05ebc10219aecbbf743214621
src/ui/fonts/inter_medium_20.c	109924	6d2067839ae61ff2ceb6c4b85da0025bc8baffe5462486e46bf555412769c08e
src/ui/fonts/inter_medium_32.c	218261	9445298cfc9ce4141e95c867e65aaf87a5cd2a5c6d6ae71ffae4b0689c8f9fe3
src/ui/fonts/inter_regular_24.c	130640	0f898381eda14ad4bf04eb3d1eca09bae336e85b0c35565ca06a719653ef0ac4
src/ui/fonts/inter_regular_32.c	195082	14680f1a8e90933a0d6d84e0e8760548cc3a456392c5703a3dc9de8053f1fd18
src/ui/fonts/inter_regular_48.c	416631	fa6169cb2efdf02dd8ba3dc4642001d1e5efcc24ede28723902003fc653c97d1
src/ui/fonts/lvgl.h	391	11c17c628828fa80cf4d76e36e738c309ee0e5bd18344dd8175625733c54ac68
src/ui/fonts/lvgl_font_compat.c	3435	67a8045562eb177d9798515529acd8c3927cd7045953c9960e8b7c35a1c991c2
src/ui/fonts/monogram_16.c	16168	a0e5970175aaeabbac7299cadbb545920af1f31d076539c44d0ff0bc4e7649a3
src/ui/fonts/monogram_16.h	158	9c1f589005022ab443d2e0320d282e4aa4c5799b3d255fab19a8df68d2d4c426
src/ui/fonts/monogram_16_bootloader.c	4840	5de1f46de5c48efdb1ee97ddde36b48ab68c1dd84ed87d92795c461a71c77fc4

src/ui/fonts/monogram_16_bootloader.h	191	be9c3f187d51c4c9d06de25ef871450fec51b85ee964da350ec7babc30f64d67	
src/ui/fonts/password_12.c	1405	3e9db0e36887259c555fe1ba6393ef9efa4f66c8f7bcb9daca4d13f66fbb724e	
src/ui/fonts/password_12.h	198	c52f920e705997665f8d6a6a14cf4071f8953dae7f047ee8781bf40b29898369	
src/ui/fonts/password_9.c	1388	b089160d8076d8b1b0fec785e8ad9e17870aaaebd543a44ce40598a0f9cda800	
src/ui/fonts/password_9.h	154	90f8f69a18d7dc70b40620af8974468d44b61f899f6db6139ba61f0f4676910b	
src/ui/fonts/regular_11.c	32527	7b364b0fc185ca3f17222133fea1ce2ea7551077cdabdeb1e9a404f7f9f1d4d5	
src/ui/fonts/regular_11.h	154	0a98c23bac34f471d61acf4fa92a2c837e07806433ca95aa131eb5086c6b8835	
src/ui/fonts/regular_12.c	33621	061359ab0984d07b83e698f7cec0082d786731378f288b383fd8710f418a946f	
src/ui/fonts/regular_12.h	154	c284235a85a0a20cbe2546d488d6588be37e6df1b84e6d780d293f646813cb9a	
src/ui/fonts/regular_9.c	31381	7aaab51be31de788729c9fe46ec4024079fe1bbf7fb5f389c064af8e0eb91a31	
src/ui/fonts/regular_9.h	150	98f7215d5be8c70c3cc3aa8e97184a5f391201367c8b73d7a3ccc56b1f528af3	
src/ui/fonts/regular_9_bootloader.c	15876	5453ccee1557b14014c4b4a7ab57004ab4b814d98f454bd04756ae80a7b2dae2	
src/ui/fonts/regular_9_bootloader.h	183	e1d321850785b1cf43e041776fb84f961f042c34baf1809870e1a716ad1483e6	
src/ui/fonts/regular_fonts.h	197	8a8e025aa12cff224593361e93ac0b80822f17945e297ba5271c0cbbbff12580	
src/ui/graphics/graphics.c	700	04b7f519cfb183a24fe900af85577a43332d093b5602cab775604fe2979e2920	
src/ui/graphics/graphics.h	496	a4ce7700e842555679eb3e4413370cefa48cd2a27861e272f5e0c0bd64fbba96	
src/ui/oled/oled.c	4379	ed7a0ba91c16163d3b5b8cd5dbba05d155c6cc8f2a46d812775324ab03a88757	
src/ui/oled/oled.h	2770	11bc1ea771b107fc399ab5830fc6a542d1b85f8f41dad91abe2d870bcf2ec96c	
src/ui/oled/oled_writer.c	1290	1e49807ebed8391f58f225bc0e3ddbc5bc47a8be969e00ffb18f8aea69c96840	
src/ui/oled/oled_writer.h	436	484da60c1c4ace43f60680ab6067265cb3201962065648c3475b8a2f04be65c4	
src/ui/oled/sh1107.c	4742	4cdbdd49fb69f9a9ae925988c4bbfc477e61fd884492347a09375e665e1de17c	
src/ui/oled/sh1107.h	424	6f11af7cd69de4fccf04a5ca441b9d39c7b4c40f06bc9937ec17283af768348c	
src/ui/oled/ssd1312.c	4806	c229caef4b0b9f99872bdf2b6015d38a180a85f9aa248521c490262ca2fb1173	
src/ui/oled/ssd1312.h	412	7e80fa18a1ff73103d7420dc2f8e028b1746d879739f49dbcc264c27821e6b9a	
src/ui/screen_process.c	2920	c797a0a0dd7f8a7519bcc53fcefb344d5b5fb78baf2691d3c40e655197e9107a	
src/ui/screen_process.h	1324	4925324ae565ea56f3b245de7e65211ba042e42d3ca1f08f9959f59a26302bbf	
src/ui/screen_saver.c	1605	e5d01795a1d6c770a03cd378030fb5badf2394bd018ed5b839dd4913f299ef7e	
src/ui/screen_saver.h	841	6d9596bf239ce5933c80ceaaf9ea6f1e8072a6abb18ced3780aa21d32ac4628f	
src/ui/screen_stack.c	2027	197e7ea15e96db7fd131ea244c960a3d4832517ebd3f0a3d79d3a64af3042241	
src/ui/screen_stack.h	541	15906232878e5958c681a35ed055844d98b3dd1aac15978929c8736e47fcb6ae	
src/ui/ugui/ugui.c	24444	220bfa84f9ce74facbe56797bb218cb57d22c4bf009f083cf8a3dc5cd03a695f	
src/ui/ugui/ugui.h	5912	d6faaeecf2aa46263906f5ac5a062b10e74583c75a783dd0f3ebee5cc98627d0	

src/ui/ui_util.c	8629	9de60147f2cb3098e19bfff7bca0b9d88c5aa3a48e1f004ea14f57d02dd4aae4
src/ui/ui_util.h	5096	c27d10d4029d2bb1093a8a9c119157e89d60f49a3d3b9c78c7f896525767a2f2
src/usb/class/hid/hid.c	11109	bcd4ce1b19d1124edb7e01ef42f03e770287cf5d1657bc6d20775a084f958019
src/usb/class/hid/hid.h	4109	577de80520fb227d0c59d54cbee8b7e2a433255b9d57b63e02de85666d087e0f
src/usb/class/hid/hww/hid_hww.c	4754	6386782800379810d5b871a51d656181425832582fe7186513a0d7cd894bc042
src/usb/class/hid/hww/hid_hww.h	1709	c55cc67135f87e9364a8b27a1cbc8c9c1c1c532c1f483357194be370fb74fb3c
src/usb/class/hid/u2f/hid_u2f.c	5262	ce8506310c076ae0cb615c00b78afab8d9958f9cbc5de15cfc7e7bd32ab55cb5
src/usb/class/hid/u2f/hid_u2f.h	1623	bcec43c5c70c6eefbbdc8d1e6005860562c9af74c9bda31084b76fd16266ec5d
src/usb/class/usb_desc.h	7046	5065da9afd04d585483cb4bb706a6ecfd53e3504c5a96c757959f36385d9c377
src/usb/class/usb_desc_bitbox02plus.h	5016	4538dd98aad944ad1a79ef1004234c299475cb6fe215ad3f4bc62b10a294172b
src/usb/class/usb_desc_common.h	6509	f81e45537c09af04b33ff8a5705458630940f4ab89805dad0b68b2f5567ea801
src/usb/class/usb_size.h	166	f9ec0deac44a546890d03fd364d6f7c30337cf3af0bf10b8dfd544210082621d
src/usb/class/usb_u2f_desc.h	3758	6d7dd70956c39337cceb388e23a971840d5d4dd1e6aac0907fb2184b52fa1b50
src/usb/u2f/u2f.h	3133	d76b09a8fb63da8ecf6b6490b165c20005247b9b1a0c391070c60e030a42b63e
src/usb/u2f/u2f_hid.h	2982	66976af1f1a767ae9c7e4632d530cfbe0e7849e4acdeb2c13d020dede7ecee9c
src/usb/u2f/u2f_keys.h	2313	a162f73d950ca857744d864a07e27d90dd9f78c0be9ddc86114a2d1ae22ff97b
src/usb/usb.c	3090	0d427e3fb7ab2dbba2b65367e70ae6d2017d9c8ff3191109467c9edb7890564e
src/usb/usb.h	275	f40df9cad0283326621b13dae6dd0027f43acb95ce640f28c7e6914a8d2fa809
src/usb/usb_frame.c	5540	92d3fe3c443ca9989db2f96c976a931517c41394342bd98f314d1d3c215c78b0
src/usb/usb_frame.h	4220	b4e0030db88b0911f7ee8a8ffa623a6bd470b0eb543eb7b4422ce6b507d3737d
src/usb/usb_packet.c	3034	a08a57316ff2a59ced1b13b2518bfc12ba116b19feede5d5e147e73a8fab2db1
src/usb/usb_packet.h	865	7d2d02d5d55c31764bd99c5585730cbbf2baf6442fcd63ccae826284baee801a
src/usb/usb_processing.c	13698	e62fb4705df80476c55490f83bd3889603916eca52be3a6cd3a80ca7794e4645
src/usb/usb_processing.h	2167	b13ab589a15524572845e28b92af5bcd1757e9bad4fa92d7e74d5a75c1fe1e27
src/util.c	3410	c89f4e11e0a50684fd7986a85a7d84749b53f7870b6e89c7d88326f2c3192452
src/util.h	5262	8bfcd192ef4b5709bdc647646e9e595c2aefd6dbcd94fa79a84c13161992183b